Diagonal Simplicial Tensor Modules and Algebraic n-Hypergroupoids

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Abstract

Let A be a commutative ring, let $k \in \mathbb{Z}^+$, and let $\vec{s} = (n_1, \ldots, n_k) \in (\mathbb{Z}^+)^k$ with $n = \min_a(n_a) - 1$. We attach to $\vec{s}$ a diagonal simplicial tensor module $X_\bullet(\vec{s}; A)$ whose p-simplices are functions on a cosimplicial index set $I_p(\vec{s}) \subseteq \mathbb{N}^k$. This extends Quillen's diagonal on double semi-simplicial groups: $X_\bullet(\vec{s}; A)$ is obtained by restricting a k-fold simplicial A-module along the diagonal $p \mapsto (p, \ldots, p)$.

Using a "missing indices" description of face kernels, we compute the horn kernels $R_{p,j}(X)$ and show that $R_{p,j}(X) \neq 0$ if and only if $k \geq p$, independently of j. Consequently, $X_\bullet(\vec{s}; A)$ is an algebraic n-hypergroupoid in the sense of Duskin (1979) and Glenn (1982) if and only if $k \leq n$, and horn fillers in dimension n are non-unique if and only if $k \geq n$; in particular it is strict precisely when $k = n$. A Horn Non-Degeneracy Lemma shows that, for $p \geq 1$, $R_{p,j}(X) \cap D_p(X) = \{0\}$ and yields a decomposition $X_p = R_{p,j}(X) \oplus D_p(X)$. An explicit shift-and-truncate chain homotopy, equivariant under $\mathrm{Stab}(\vec{s})$ and compatible with a natural filtration, contracts $X_\bullet(\vec{s}; A)$ and forces the associated spectral sequence to collapse at E_1.

When A is an infinite field K, we study simplicial submodules generated by a single tensor via kernel sequences and a moduli map to a product of Grassmannians. The moduli map image is an irreducible and unirational constructible subset of a determinantal incidence variety. Finally, we extend the construction by fixed rank shifts. We compute the homology of the shifted ambient modules, embed the simplicial chain module of every finite graph in an order-three example, and show that reduced representations are closed under smash products by concatenating tensor axes. Sparse rational computations illustrate the Petersen graph and the smash product of two simplicial circles.

Keywords: diagonal simplicial tensor modules, horn kernels, strict algebraic n-hypergroupoids, incidence varieties, moduli spaces

2020 MSC: Primary 18G30; Secondary 55U10, 14M15, 05E45, 15A69.

1. Introduction

Let A be a commutative ring and $k \in \mathbb{Z}^+$. Fix a **shape** $\vec{s} = (n_1, \ldots, n_k) \in (\mathbb{Z}^+)^k$ and set $n = \min(\vec{s}) - 1$. We construct the diagonal simplicial tensor module $X_\bullet(\vec{s}; A)$ (DSTM), a simplicial A-module whose p-simplices are functions on cosimplicial index sets $I_p(\vec{s}) \subseteq \mathbb{N}^k$.

This construction generalizes Quillen's diagonal on a double semi-simplicial group [1]: it is the restriction $p \mapsto (p, \ldots, p)$ of a k-fold simplicial A-module in $\mathbf{Mod}_A$ (Remark 2.3), and is compatible with Dold–Kan normalization [2, 3, 4].

In particular, the diagonal simplicial module $X_\bullet(\vec{s}; A)$ is obtained from $X_\bullet(\vec{s}; \mathbb{Z})$ by base change, and the normalized Moore complexes satisfy

$$N_\bullet\big(X_\bullet(\vec{s}; A)\big) \cong N_\bullet\big(X_\bullet(\vec{s}; \mathbb{Z})\big) \otimes_{\mathbb{Z}} A$$

for all commutative rings A.

We analyze the horn kernels

$$R_{p,j}(X) = \bigcap_{i \neq j} \ker\big(d_i : X_p \to X_{p-1}\big),$$

and show that their combinatorics controls the homotopy-theoretic behavior of the DSTM. For the simplicial dimension $n = \min(\vec{s}) - 1$ we prove that $X_\bullet(\vec{s}; A)$ is a strict algebraic n-hypergroupoid in this sense if and only if the tensor order satisfies $k = n$. The transition from non-unique to unique horn fillers occurs at $p = k$ (hence at $p = n$ in the strict case $k = n$).

We briefly outline the main aspects of the paper: combinatorics and classification of the DSTM, and the algebraic geometry of generated submodules.

Combinatorics and Classification

In the abelian setting of simplicial modules, horn fillers always exist (the Kan condition is automatic). The obstruction to the uniqueness of fillers is captured by the horn kernel $R_{p,j}$. Following terminology from Duskin and Glenn [5, 6], a simplicial module is a **strict algebraic n-hypergroupoid** if fillers are unique in dimensions strictly greater than n ($R_{p,j} = 0$ for $p > n$), and non-unique in dimension n ($R_{n,j} \neq 0$).

We characterize $R_{n,j}$ combinatorially via a basis of "missing indices." The Horn Non-Degeneracy Lemma proves $R_{n,j} \cap D_n = \{0\}$, where D_n denotes the degenerate submodule in degree n, yielding the decomposition $X_n = R_{n,j} \oplus D_n$. Counting missing indices gives exact rank formulas for $R_{p,j}$ via inclusion–exclusion on the product sets $I_p = \prod_a [M_a(p)]$. In the constant shape case $n_a = n + 1$ for all a, so $M_a(p) = p$ and $I_p = [p]^k$, and these rank formulas specialize to finite differences of x^k and can be expressed using Stirling numbers of the second kind $\left\{ {k \atop m} \right\}$ [7]. By analyzing when the horn kernel vanishes, we obtain our main classification theorem: $X_\bullet(\vec{s}; A)$ is a strict algebraic n-hypergroupoid if and only if $k = n$.

Algebraic Geometry of Generated Submodules

When A is an infinite field K, we work throughout with submodules generated by a single tensor $T \in X_n(\vec{s}; K)$; we do not attempt to parametrize arbitrary subcomplexes. Such a generated submodule $\langle T \rangle$ is encoded by its kernel sequence

$$K(T)_\bullet := (\ker f_{T,p})_p \subseteq K[\mathbf{\Delta}^n]_\bullet,$$

where f_T is the realization map from Section 8.

We define a moduli map Ψ from a Zariski-open locus $\mathcal{U} \subset X_n(\vec{s}; K)$ to a product of Grassmannians by sending T to the collection of subspaces $K(T)_p \subseteq K[\mathbf{\Delta}^n]_p$. The image $\mathcal{M}(\vec{s})$ lies within a closed incidence subvariety defined by simplicial compatibility conditions, which we show are linear in the Segre–Plücker coordinates. This reduces the classification of these generated submodules to the geometry of these incidence conditions and of the index collision maps $\mathcal{I}_p$.

Rank shifts and homology representations

Fixed rank shifts enlarge the tensor axes without changing the number of simplicial operators in each degree. Although the resulting ambient modules have zero homology in positive degrees, their simplicial submodules can represent nontrivial chain complexes. We give an explicit order-three representation of every finite graph: vertices and normalized edges remain linearly independent, and the normalized differential is the incidence map. Tensor products concatenate axes and yield representations of reduced smash products. The Petersen graph and the smash of two circles provide examples computed exactly over the rationals by the accompanying sparse implementation.

Organization

Section 2 establishes notation and the DSTM construction. Section 3 characterizes missing indices and the horn kernel basis. Section 4 develops the Moore filler algorithm and normalization. Section 5 establishes rank formulas and the classification theorem. Section 6 proves the Horn Non-Degeneracy Lemma and the decomposition $X_n = R_{n,j} \oplus D_n$. Section 7 develops the homology dichotomy and the hypergroupoid classification. Section 8 develops the algebraic geometry of generated submodules. Section 9 treats rank shifts, graph representations, and reduced smash products. In Appendix A we prove contractibility via an explicit equivariant homotopy.

2. Diagonal simplicial tensor modules

Throughout, let A be a commutative ring and write $\mathbf{Mod}_A$ for the category of A-modules. Denote by $\mathbb{N}$ the set of nonnegative integers and by $\mathbb{Z}^+$ the set of positive integers.

2.1. Simplicial preliminaries

The **simplicial category** $\mathbf{\Delta}$ has objects $[p] := \{0, \dots, p\}$ for $p \in \mathbb{N}$; morphisms are nondecreasing maps. It is generated by the **coface maps** $\delta_i^p : [p-1] \to [p]$ (the injection missing i) and the **codegeneracy maps** $\sigma_i^p : [p+1] \to [p]$ (the surjection repeating i).

A **simplicial A-module** is a functor $X_\bullet : \mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Mod}_A$. We write $X_p := X([p])$. The induced maps are the **face maps** $d_i^p := X(\delta_i^p) : X_p \to X_{p-1}$ and the **degeneracy maps** $s_i^p := X(\sigma_i^p) : X_p \to X_{p+1}$. The associated chain complex $(X_\bullet, \partial_\bullet)$ has boundary maps $\partial_p = \sum_{i=0}^{p} (-1)^i d_i^p$.

2.2. Setup and index sets

We now define the index sets underlying the diagonal simplicial tensor module. Let $k \in \mathbb{Z}^+$. We analyze modules consisting of tensors of order k. A **shape** is a k-tuple $\vec{s} = (n_1, \dots, n_k) \in (\mathbb{Z}^+)^k$.

We fix the shape $\vec{s}$ throughout and define its **simplicial dimension** as

$$n = n(\vec{s}) := \min(n_1, \dots, n_k) - 1.$$

For any degree $p \geq 0$ and axis $1 \leq a \leq k$, we define the index bounds:

$$M_a(p) := n_a - 1 - n + p.$$

Since $n_a - 1 \geq n$, we have $M_a(p) \geq p$ for all a. The index set in degree p is defined as

$$I_p := \prod_{a=1}^{k} [\, M_a(p) \,].$$

Since $[p] \subseteq [\, M_a(p) \,]$ for every a, we have the inclusion $[p]^k \subseteq I_p$.

2.3. The Cosimplicial Index Set $I_\bullet$

We define a functor $I_\bullet : \mathbf{\Delta} \to \mathbf{Set}$. On objects, $I_\bullet([p]) := I_p$. Note that $M_a(p \pm 1) = M_a(p) \pm 1$.

The structure maps are defined by applying the same simplicial generator in each tensor axis.

Definition 2.1 (Cosimplicial Structure Maps)**.** The functor $I_\bullet$ maps the generators of $\mathbf{\Delta}$ as follows:

For $p \geq 1$ and $0 \leq i \leq p$, the **index coface map** $\Delta_i^p : I_{p-1} \to I_p$ is

$$\Delta_i^p := I_\bullet(\delta_i^p) := \prod_{a=1}^{k} \delta_i^{M_a(p)}.$$

For $p \geq 0$ and $0 \leq i \leq p$, the **index codegeneracy map** $\Sigma_i^p : I_{p+1} \to I_p$ is

$$\Sigma_i^p := I_\bullet(\sigma_i^p) := \prod_{a=1}^{k} \sigma_i^{M_a(p)}.$$

Since $M_a(p) \geq p$, all factors are defined for the common index i. Because the standard maps δ_i, σ_i satisfy the cosimplicial identities, the product maps Δ_i^p, Σ_i^p also satisfy them componentwise, proving that $I_\bullet$ is a cosimplicial set.

2.4. The Diagonal Simplicial Module $X_\bullet(\vec{s}; A)$

We define the contravariant functor $A^{(-)} : \mathbf{Set} \to \mathbf{Mod}_A$. This functor maps a set S to the A-module of functions $S \to A$, and a map of sets $f : S_1 \to S_2$ to the module homomorphism $A^f : A^{S_2} \to A^{S_1}$ defined by pre-composition $(T \mapsto T \circ f)$.

Definition 2.2 (Diagonal simplicial tensor module)**.** The **diagonal simplicial A-tensor module** $X_\bullet(\vec{s}; A) : \mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Mod}_A$ is the composition $A^{(-)} \circ I_\bullet$.

On objects, $X_p(\vec{s}; A) := A^{I_p}$. Since the index sets I_p are finite, X_p is a free A-module.

The face and degeneracy maps are induced by the corresponding maps in $I_\bullet$ via pre-composition:

$$d_i^p(T) := T \circ \Delta_i^p,$$
$$s_i^p(T) := T \circ \Sigma_i^p.$$

Remark 2.3. Define $\widetilde{X} : (\mathbf{\Delta}^{\mathrm{op}})^k \to \mathbf{Mod}_A$ by

$$\widetilde{X}_{p_1,\dots,p_k} := A^{\prod_{a=1}^{k}[M_a(p_a)]}.$$

In the a-th simplicial direction, the face and degeneracy maps are induced by precomposition with $\delta_i^{M_a(p_a)}$ and $\sigma_i^{M_a(p_a)}$ on the a-th factor (identity on the others). Then

$$X_p(\vec{s}; A) = \widetilde{X}_{p,\dots,p},$$

so the structure maps of $X_\bullet(\vec{s}; A)$ are obtained by applying the same simplicial operator in each direction.

Remark 2.4 (Matrices in the constant shape case). If $k = 2$ and $\vec{s} = (N, N)$, then $n = N - 1$ and

$$M_a(p) = n_a - 1 - n + p = p \qquad (a = 1, 2).$$

Hence

$$I_p = [p]^2 \qquad \text{and} \qquad X_p(\vec{s}; A) = A^{[p]^2} \cong A^{(p+1)\times(p+1)}.$$

In particular,

$$X_{N-1}(\vec{s}; A) \cong A^{[N-1]^2} \cong A^{N\times N},$$

so an $N \times N$ matrix may be viewed as a simplex in degree $N - 1$ of $X_\bullet(\vec{s}; A)$.

2.5. Axis Symmetries and Permutations of Tensor Factors

For a shape $\vec{s} = (n_1, \dots, n_k)$, define its stabilizer

$$\mathrm{Stab}(\vec{s}) := \{\sigma \in S_k : n_{\sigma(a)} = n_a \text{ for all } a\}.$$

Each $\sigma \in \mathrm{Stab}(\vec{s})$ acts (on the left) on the index sets I_p by permuting coordinates:

$$\sigma \cdot (m_1, \dots, m_k) := (m_{\sigma^{-1}(1)}, \dots, m_{\sigma^{-1}(k)}).$$

This induces a (left) action on the diagonal simplicial module $X_\bullet(\vec{s}; A)$ by precomposition:

$$(\sigma \cdot T)(m) := T(\sigma^{-1} \cdot m), \qquad T \in X_p(\vec{s}; A),\ m \in I_p.$$

Lemma 2.5 (Equivariance under $\mathrm{Stab}(\vec{s})$)**.** *The action of* $\mathrm{Stab}(\vec{s})$ *on* $I_\bullet$ *commutes with the cosimplicial structure maps. Consequently, for every* $\sigma \in \mathrm{Stab}(\vec{s})$ *and all* p, i *we have*

$$d_i^p(\sigma \cdot T) = \sigma \cdot d_i^p(T), \qquad s_i^p(\sigma \cdot T) = \sigma \cdot s_i^p(T).$$

The boundary operators ∂_p *also commute with the* $\mathrm{Stab}(\vec{s})$*-action.*

Proof. Let $\sigma \in \mathrm{Stab}(\vec{s})$. By definition, $n_a = n_{\sigma(a)}$, which implies $n_a = n_{\sigma^{-1}(a)}$. This ensures $M_a(p) = M_{\sigma^{-1}(a)}(p)$ for all a, p.

We first verify the commutation on the index sets. Let $m \in I_{p-1}$. We show $\sigma \cdot \Delta_i^p(m) = \Delta_i^p(\sigma \cdot m)$. The a-th component of the left side is

$$(\sigma \cdot \Delta_i^p(m))_a = (\Delta_i^p(m))_{\sigma^{-1}(a)} = \delta_i^{M_{\sigma^{-1}(a)}(p)}(m_{\sigma^{-1}(a)}).$$

The a-th component of the right side is

$$(\Delta_i^p(\sigma \cdot m))_a = \delta_i^{M_a(p)}((\sigma \cdot m)_a) = \delta_i^{M_a(p)}(m_{\sigma^{-1}(a)}).$$

Since $M_a(p) = M_{\sigma^{-1}(a)}(p)$, the components are equal. Thus $\sigma \circ \Delta_i^p = \Delta_i^p \circ \sigma$. The argument for Σ_i^p is analogous.

Now we verify the equivariance on X_p. Let $T \in X_p$.

$$\begin{aligned} d_i^p(\sigma \cdot T) &= (\sigma \cdot T) \circ \Delta_i^p \\ &= (T \circ \sigma^{-1}) \circ \Delta_i^p \\ &= T \circ (\sigma^{-1} \circ \Delta_i^p) \\ &= T \circ (\Delta_i^p \circ \sigma^{-1}) \quad \text{(by commutativity on } I_\bullet) \\ &= (T \circ \Delta_i^p) \circ \sigma^{-1} \\ &= \sigma \cdot (T \circ \Delta_i^p) = \sigma \cdot d_i^p(T). \end{aligned}$$

The proof for s_i^p is analogous. The statement for ∂_p follows by linearity. □

Remark 2.6 (Constant shape and symmetry subcomplexes). Suppose $\vec{s} = (N, \dots, N)$ is constant. Then $\mathrm{Stab}(\vec{s}) \cong S_k$ acts on $X_\bullet(\vec{s}; A)$ by permuting tensor axes. By Lemma 2.5, the symmetric and alternating subspaces defined below are preserved by all structure maps and hence form simplicial A-submodules.

In particular, the symmetric and alternating subspaces

$$X_\bullet^{\mathrm{Sym}} := \{T : \sigma \cdot T = T \ \forall \sigma \in S_k\}, \qquad X_\bullet^{\mathrm{Alt}} := \{T : \sigma \cdot T = \mathrm{sgn}(\sigma)\, T\}$$

form simplicial A-submodules. (If $k!$ is invertible in A, these are direct summands.) In Appendix A we show that the global contracting homotopy H is $\mathrm{Stab}(\vec{s})$-equivariant, so it restricts to chain contractions on $X_\bullet^{\mathrm{Sym}}$ and $X_\bullet^{\mathrm{Alt}}$. Hence these symmetry subcomplexes are also contractible.

Remark 2.7 (Matrix and Hermitian transpose). For $k = 2$ and $\vec{s} = (N, N)$, the nontrivial element $\tau \in S_2$ acts by swapping the tensor axes, corresponding to the transpose: $(\tau \cdot T)(m_1, m_2) = T(m_2, m_1)$. By Lemma 2.5, all structure maps and the boundary operators ∂_p commute with this action.

Over $K = \mathbb{C}$, viewing $X_\bullet(\vec{s}; K)$ as a simplicial $\mathbb{R}$-module, complex conjugation commutes with all structure maps. The Hermitian conjugate action (transpose combined with conjugation) also commutes with d_i^p, s_i^p and ∂_p (as $\mathbb{R}$-linear operators). The real subspaces of symmetric, skew-symmetric, Hermitian, and skew-Hermitian tensors form simplicial $\mathbb{R}$-submodules, and the global contracting homotopy H restricts to each of them.

2.6. Base Change and Realization Maps

Proposition 2.8 (Base change for $X_\bullet(\vec{s}; -)$)**.** *For any ring homomorphism $\varphi : A \to B$ and shape $\vec{s}$, there is a canonical isomorphism of simplicial B-modules*

$$X_\bullet(\vec{s}; A) \otimes_A B \xrightarrow[\cong]{\theta_\bullet} X_\bullet(\vec{s}; B).$$

Proof. Each $X_p(\vec{s}; A) = A^{I_p}$ is a finite free A-module with basis $\{E_m\}_{m \in I_p}$ (indicator functions). Define

$$\theta_p : X_p(\vec{s}; A) \otimes_A B \longrightarrow X_p(\vec{s}; B), \qquad \theta_p(E_m \otimes b) = b \cdot E_m.$$

Since I_p is finite, $(A^{I_p}) \otimes_A B \cong B^{I_p}$, so each θ_p is a B-module isomorphism.

Simplicial compatibility follows because the face and degeneracy maps are given by precomposition with the same maps Δ_i^p, Σ_i^p on $I_\bullet$ in both $X_\bullet(\vec{s}; A)$ and $X_\bullet(\vec{s}; B)$, so each θ_p commutes with all d_i^p and s_i^p. □

Corollary 2.9 (Naturality and flat base change)**.** *Let $T \in X_n(\vec{s}; A)$, and let $f_T : A[\mathbf{\Delta}^n] \to X_\bullet(\vec{s}; A)$ be the corresponding realization map (defined in Section 8). Let $\varphi : A \to B$ be a ring homomorphism, and let $T_B := \theta_n(T \otimes 1) \in X_n(\vec{s}; B)$. Let $\varphi_p : A[\mathbf{\Delta}^n]_p \otimes_A B \xrightarrow{\cong} B[\mathbf{\Delta}^n]_p$ denote the canonical isomorphism.*

1. *(Naturality) The following diagram commutes for all p:*

$$\begin{array}{ccc} A[\mathbf{\Delta}^n]_p \otimes_A B & \xrightarrow{f_{T,p} \otimes 1_B} & X_p(\vec{s}; A) \otimes_A B \\ {\scriptstyle \varphi_p}\downarrow{\scriptstyle \cong} & & {\scriptstyle \theta_p}\downarrow{\scriptstyle \cong} \\ B[\mathbf{\Delta}^n]_p & \xrightarrow[f_{T_B,p}]{} & X_p(\vec{s}; B) \end{array}$$

2. *(Kernel preservation) If B is a flat A-module, the kernel sequence is preserved:*

$$(\ker f_{T,p}) \otimes_A B \;\cong\; \ker(f_{T_B,p}) \qquad (\forall p).$$

Proof. (1) Let $\alpha \in \mathbf{\Delta}([p], [n])$ be a basis element of $A[\mathbf{\Delta}^n]_p$. Both paths in the diagram map the element $\alpha \otimes b$ to $b \cdot f_{T_B,p}(\alpha)$, establishing commutativity.

(2) Consider the exact sequence $0 \to \ker f_{T,p} \to A[\mathbf{\Delta}^n]_p \xrightarrow{f_{T,p}} X_p(\vec{s}; A)$. If B is a flat A-module, the functor $(-) \otimes_A B$ is exact. Applying it yields the exact sequence

$$0 \to (\ker f_{T,p}) \otimes_A B \to A[\mathbf{\Delta}^n]_p \otimes_A B \xrightarrow{f_{T,p} \otimes 1_B} X_p(\vec{s}; A) \otimes_A B.$$

This implies $(\ker f_{T,p}) \otimes_A B \cong \ker(f_{T,p} \otimes 1_B)$. By the commutativity of the diagram in (1) and the fact that φ_p and θ_p are isomorphisms, $\ker(f_{T,p} \otimes 1_B)$ is isomorphic to $\ker(f_{T_B,p})$. □

3. Horns, Kernels, and Missing Indices

We investigate the structure of face kernels and the uniqueness of fillers for horns in $X_\bullet(\vec{s}; A)$.

3.1. Horns and the Horn Kernel

Let $p \geq 1$ and $0 \leq j \leq p$. We denote $F_j = [p] \setminus \{j\}$.

Definition 3.1 ((p,j)-Horn and Filler)**.** A (p,j)**-horn** in $X_\bullet$ is a tuple $H = (h_i)_{i\in F_j}$ of elements $h_i \in X_{p-1}$ satisfying the **horn compatibility condition**:

$$d_i(h_\ell) = d_{\ell-1}(h_i) \quad \text{for all } i < \ell \text{ with } i,\ell \in F_j.$$

A tensor $T \in X_p$ is a **filler** of H if $d_i(T) = h_i$ for all $i \in F_j$.

In the category of simplicial modules, every horn has at least one filler. The uniqueness of fillers is measured by the horn kernel.

Definition 3.2 (Horn Kernel)**.** The (p,j)**-horn kernel** $R_{p,j}$ is the submodule of X_p defined by

$$R_{p,j} := \bigcap_{i\in F_j} \ker(d_i^p).$$

If T and T' are two fillers for the same horn H, their difference $T - T' \in R_{p,j}$. The set of fillers $\mathcal{F}(H)$ is an affine space (torsor) modeled on $R_{p,j}$. Fillers are unique if and only if $R_{p,j} = \{0\}$.

3.2. Structure of the Face Kernels

We characterize the kernels of the face maps using the standard basis $\{E_m\}_{m\in I_p}$ of X_p.

Lemma 3.3 (Action on basis elements)**.** *Let $m \in I_p$. The action of the face map d_i^p on E_m is*

$$d_i^p(E_m) = \begin{cases} E_{m'} & \text{if } m \in \operatorname{im}(\Delta_i^p), \text{ where } m' \text{ is the unique preimage,} \\ 0 & \text{if } m \notin \operatorname{im}(\Delta_i^p). \end{cases}$$

Proof. Evaluate at $m'' \in I_{p-1}$: $(d_i^p(E_m))(m'') = E_m(\Delta_i^p(m'')) = \delta_{m,\Delta_i^p(m'')}$. If $m \notin \operatorname{im}(\Delta_i^p)$, this is zero. If $m \in \operatorname{im}(\Delta_i^p)$, since Δ_i^p is injective, there is a unique m' such that $\Delta_i^p(m') = m$. Then $d_i^p(E_m) = E_{m'}$. □

Lemma 3.4 (Injectivity Lemma)**.** *If $d_i^p(E_{m_1}) = d_i^p(E_{m_2}) \neq 0$, then $m_1 = m_2$.*

Proof. By Lemma 3.3, the common value is some $E_{m^*} \in X_{p-1}$. Then $m_1 = \Delta_i^p(m^*)$ and $m_2 = \Delta_i^p(m^*)$. □

Theorem 3.5 (Basis of the Face Kernel)**.** *The kernel of $d_i^p : X_p \to X_{p-1}$ is the free A-submodule*

$$\ker(d_i^p) = \operatorname{span}_A\{E_m \mid m \in I_p \setminus \operatorname{im}(\Delta_i^p)\}.$$

Proof. Let $T = \sum_{m\in I_p} a_m E_m \in X_p$. Applying d_i^p gives

$$d_i^p(T) = \sum_{m\in\operatorname{im}(\Delta_i^p)} a_m d_i^p(E_m),$$

since $d_i^p(E_m) = 0$ if $m \notin \operatorname{im}(\Delta_i^p)$ (Lemma 3.3). By the Injectivity Lemma 3.4, the set $\{d_i^p(E_m) \mid m \in \operatorname{im}(\Delta_i^p)\}$ consists of distinct standard basis elements in X_{p-1}, hence is A-linearly independent.

Thus $d_i^p(T) = 0$ if and only if $a_m = 0$ for all $m \in \operatorname{im}(\Delta_i^p)$. Equivalently, T is an A-linear combination of the E_m with $m \notin \operatorname{im}(\Delta_i^p)$, which proves the stated basis description of $\ker(d_i^p)$. □

3.3. Missing Indices and the Support Characterization

Definition 3.6 (Missing Index)**.** An index $m \in I_p$ is **missing** from the (p, j)-horn if

$$\forall i \in F_j : m \notin \mathrm{im}(\Delta_i^p).$$

Let $M_{p,j} \subset I_p$ denote the set of missing indices.

Theorem 3.7 (Support Characterization / Horn Kernel Basis)**.** *The horn kernel $R_{p,j}$ is a free A-module with basis $\{E_m\}_{m \in M_{p,j}}$. Consequently, if T_1 and T_2 are two fillers of the same horn, their difference is supported on the missing indices:* $\mathrm{supp}(T_1 - T_2) \subseteq M_{p,j}$.

Proof. By Theorem 3.5, $\ker(d_i^p)$ is the coordinate subspace spanned by the basis $B_i = \{E_m \mid m \notin \mathrm{im}(\Delta_i^p)\}$. The horn kernel $R_{p,j}$ is the intersection of these coordinate subspaces for $i \in F_j$. Since each $\mathrm{span}_A(B_i)$ consists exactly of those tensors whose coefficients vanish outside B_i, their intersection consists of those tensors whose coefficients vanish outside $\bigcap_{i \in F_j} B_i$, i.e. it is $\mathrm{span}_A\big(\bigcap_{i \in F_j} B_i\big)$.

An element E_m lies in this intersection if and only if $m \notin \mathrm{im}(\Delta_i^p)$ for all $i \in F_j$, which defines $M_{p,j}$. The consequence regarding the support of the difference follows because $T_1 - T_2 \in R_{p,j}$. □

3.4. Combinatorial Characterization

We provide a combinatorial interpretation of missing indices. We denote the image of a multi-index $m = (m_1, \ldots, m_k)$ by $\mathrm{im}(m) = \{m_1, \ldots, m_k\}$.

Proposition 3.8 (Characterization of Missing Indices)**.** *An index $m \in I_p$ is missing from the (p, j)-horn ($m \in M_{p,j}$) if and only if* $\mathrm{im}(m) \supseteq F_j$.

Proof. We first establish the equivalence: $m \in \mathrm{im}(\Delta_i^p) \iff i \notin \mathrm{im}(m)$. Recall $\Delta_i^p = \prod_{a=1}^{k} \delta_i^{M_a(p)}$. The image of δ_i^q is $[q] \setminus \{i\}$.

$$\begin{aligned} m \in \mathrm{im}(\Delta_i^p) &\iff \forall a : m_a \in \mathrm{im}(\delta_i^{M_a(p)}) \\ &\iff \forall a : m_a \neq i \quad (\text{since } i \in [p] \subseteq [M_a(p)]) \\ &\iff i \notin \mathrm{im}(m). \end{aligned}$$

Taking the contrapositive, $m \notin \mathrm{im}(\Delta_i^p) \iff i \in \mathrm{im}(m)$.

By definition, $m \in M_{p,j}$ if and only if $\forall i \in F_j : m \notin \mathrm{im}(\Delta_i^p)$. Therefore,

$$m \in M_{p,j} \iff \forall i \in F_j : i \in \mathrm{im}(m) \iff F_j \subseteq \mathrm{im}(m).$$

□

Corollary 3.9. *The horn kernel is non-trivial ($R_{p,j} \neq \{0\}$) if and only if $k \geq p$.*

Proof. By Theorem 3.7, $R_{p,j} \neq \{0\}$ iff $M_{p,j} \neq \emptyset$.

($\Rightarrow$) If $m \in M_{p,j}$, then $k \geq |\mathrm{im}(m)| \geq |F_j| = p$.

($\Leftarrow$) Suppose $k \geq p$. Let $F_j = \{f_1, \ldots, f_p\}$. Set

$$m := (f_1, \ldots, f_p, 0, \ldots, 0) \in [p]^k \subseteq I_p.$$

Then $\mathrm{im}(m) \supseteq F_j$, so $m \in M_{p,j}$. □

3.5. The ℓ-free Subspace

Fix a degree $p \geq 1$ and an index $0 \leq \ell \leq p$. The combinatorial characterization of $\mathrm{im}(\Delta_\ell^p)$ yields a structural property of the face maps.

Definition 3.10 (ℓ-free subspace)**.** Define the ℓ-free subspace

$$X_p^{(\ell\text{-free})} := \mathrm{span}_A\{\, E_m : \ \ell \notin \mathrm{im}(m) \,\} \ \subseteq X_p.$$

Corollary 3.11 (Isomorphism on the ℓ-free subspace)**.** *The face map d_ℓ^p induces a linear isomorphism*

$$d_\ell^p : \ X_p^{(\ell\text{-free})} \ \xrightarrow{\cong} \ X_{p-1}.$$

Proof. By the proof of Proposition 3.8, $m \in \mathrm{im}(\Delta_\ell^p) \iff \ell \notin \mathrm{im}(m)$. Thus $X_p^{(\ell\text{-free})}$ is spanned by $\{E_m \mid m \in \mathrm{im}(\Delta_\ell^p)\}$. By the Injectivity Lemma 3.4, d_ℓ^p is injective on this subspace. Since $\Delta_\ell^p : I_{p-1} \to \mathrm{im}(\Delta_\ell^p)$ is a bijection, Lemma 3.3 shows that d_ℓ^p maps the basis of $X_p^{(\ell\text{-free})}$ bijectively onto the basis of X_{p-1}. □

3.6. Genericity

Fix $p \geq 1$ and $0 \leq j \leq p$, and assume A is an infinite field and $k \geq p$.

For $T \in X_p$, set

$$T' := \mu_j(\Phi_j(T)),$$

the Moore filler of the horn $\Phi_j(T)$ (Section 4). Then $\Phi_j(T') = \Phi_j(T)$, hence $T - T' \in R_{p,j}$. By Theorem 3.7, $\mathrm{supp}(T - T') \subseteq M_{p,j}$.

Definition 3.12 (Generic Tensor)**.** A tensor $T \in X_p$ is called (p, j)**-generic** if the difference $T - T'$ has maximal support: $\mathrm{supp}(T - T') = M_{p,j}$. (This requires $k \geq p$).

Proposition 3.13 (Generic Locus)**.** *Assume A is an infinite field and $k \geq p$. The set of (p, j)-generic tensors forms a Zariski-open subset $\mathcal{U}_{p,j} \subset X_p$.*

Proof. Define the A-linear endomorphism

$$R := \mathrm{id}_{X_p} - \mu_j \circ \Phi_j : X_p \to X_p.$$

Then $R(T) = T - \mu_j(\Phi_j(T)) = T - T' \in R_{p,j}$. Write

$$R(T) = \sum_{m \in M_{p,j}} c_m(T)\, E_m.$$

Each coefficient $c_m(T)$ is an A-linear form in the entries of T.

We show that $c_m(T)$ is not identically zero. Consider $T = E_m$ for $m \in M_{p,j}$. Since $E_m \in R_{p,j}$, its horn is $\Phi_j(E_m) = 0$. The Moore filler of the zero horn is $T' = \mu_j(0) = 0$. Then $R(E_m) = E_m$, so $c_m(E_m) = 1$.

The generic locus is defined by the non-vanishing of these forms:

$$\mathcal{U}_{p,j} \ := \ \{\, T \in X_p : \ c_m(T) \neq 0 \text{ for all } m \in M_{p,j} \,\}.$$

The complement is the union of the zero loci of the non-zero forms c_m, hence a finite union of proper linear subspaces. Therefore $\mathcal{U}_{p,j}$ is Zariski-open. □

4. Normalization and the Moore Filler

We review the normalization of simplicial modules. Let $X_\bullet$ be a simplicial A-module.

4.1. The Normalization Theorem

Definition 4.1 (Normalized and degenerate submodules)**.** The **normalized submodule** in degree p is $N_p(X) := \bigcap_{i=1}^{p} \ker d_i \subseteq X_p$. The **degenerate submodule** in degree p is $D_p(X) := \sum_{r=0}^{p-1} \operatorname{im}(s_r) \subseteq X_p$.

Note that $N_p(X) = R_{p,0}(X)$ in the notation of Section 3.

By the normalization theorem, the homology of $X_\bullet$ is isomorphic to the homology of its normalized (Moore) complex $(N_\bullet(X), d_0)$.

Theorem 4.2 (Normalization theorem / Dold–Kan Correspondence)**.** *For every $p \geq 0$ there is a functorial direct sum decomposition*

$$X_p \cong N_p(X) \oplus D_p(X).$$

There is a functorial projection $\pi_p : X_p \to N_p(X)$ with $\ker(\pi_p) = D_p(X)$; one explicit choice is given in Remark 4.3. Moreover, X_p decomposes as a direct sum of images of $N_q(X)$ under iterated degeneracies (Eilenberg–Zilber decomposition):

$$X_p \;=\; \bigoplus_{q=0}^{p} \; \bigoplus_{0 \leq i_1 < \cdots < i_{p-q} \leq p-1} s_{i_{p-q}} \cdots s_{i_1}\, N_q(X).$$

Refs.: *Dold [2]; Kan [3]; Weibel [4, Thm 8.3.8, Cor 8.4.2].*

Remark 4.3 (Eilenberg–Mac Lane idempotents). The decomposition is established by the **Eilenberg–Mac Lane idempotents**. The projection onto $N_p(X) = \bigcap_{i=1}^{p} \ker d_i$ corresponding to the convention $(N_\bullet, d_0)$ is

$$\pi_p := (\mathrm{id} - s_0 d_1) \circ (\mathrm{id} - s_1 d_2) \circ \cdots \circ (\mathrm{id} - s_{p-1} d_p),$$

so that the d_p-correction is applied first. *Refs.:* Mac Lane [8, Ch. VIII, §6]; Weibel [4, Thm 8.3.8].

4.2. The Moore Horn Filler Algorithm

Definition 4.4 (Horn space)**.** Fix $p \geq 1$ and $j \in \{0, \ldots, p\}$. Let $F_j = [p] \setminus \{j\}$. The space of compatible (p, j)-horns, $\mathrm{Horns}(p, j)$, is the submodule of $\prod_{i \in F_j} X_{p-1}$ defined by the compatibility conditions $d_i x_\ell = d_{\ell-1} x_i$ for all $i < \ell$ with $i, \ell \in F_j$.

Define the horn map

$$\Phi_j : X_p \to \mathrm{Horns}(p, j), \qquad \Phi_j(T) = (d_i T)_{i \in F_j},$$

(viewing the tuple $(d_i T)_{i \in F_j}$ as an element of $\mathrm{Horns}(p, j)$). Its kernel is the horn kernel

$$R_{p,j}(X) = \bigcap_{i \in F_j} \ker(d_i).$$

Definition 4.5 (Moore Filler Map)**.** The **Moore filler map** $\mu_j : \mathrm{Horns}(p,j) \to D_p(X)$ is defined iteratively for a horn $H = (x_i)_{i \in F_j}$.

Phase I (Ascending indices $i < j$). Initialize $T^{(-1)} := 0$. For $i = 0, \ldots, j-1$:

$$T^{(i)} := T^{(i-1)} + s_i\big(x_i - d_i T^{(i-1)}\big).$$

Phase II (Descending indices $i > j$). Initialize $U^{(p+1)} := T^{(j-1)}$. For $i = p, \ldots, j+1$:

$$U^{(i)} := U^{(i+1)} + s_{i-1}\big(x_i - d_i U^{(i+1)}\big).$$

Output. $\mu_j(H) := U^{(j+1)}$.

By construction, $\mu_j(H) \in D_p(X)$. The map μ_j is A-linear and satisfies

$$\Phi_j \circ \mu_j = \mathrm{id}_{\mathrm{Horns}(p,j)}\,.$$

see Weibel [4, Lemma 8.2.6] or Duskin [5, Lemma 3.1].

Proposition 4.6 (Exactness of the Horn Sequence)**.** *The sequence*

$$0 \longrightarrow R_{p,j}(X) \longrightarrow X_p \xrightarrow{\Phi_j} \mathrm{Horns}(p,j) \longrightarrow 0$$

is exact.

Proof. Exactness at $R_{p,j}(X)$ and X_p is by definition. Surjectivity of Φ_j follows since μ_j is a section. □

Corollary 4.7. *The Moore filler map μ_j induces a direct sum decomposition*

$$X_p = R_{p,j}(X) \oplus \mathrm{im}(\mu_j).$$

Furthermore, $\mathrm{im}(\mu_j) \subseteq D_p(X)$.

Proof. Since μ_j is a section of Φ_j ($\Phi_j \circ \mu_j = \mathrm{id}_{\mathrm{Horns}(p,j)}$), the exact sequence in Proposition 4.6 splits, yielding the decomposition. The inclusion follows from Definition 4.5. □

5. Combinatorics and Classification

We apply the combinatorial characterization of the horn kernel (Section 3) to classify the DSTM $X_\bullet(\vec{s}; A)$ and derive exact rank formulas.

5.1. Classification Theorem

We establish the main classification result based on the tensor order k and the simplicial dimension $n = \min(\vec{s}) - 1$.

Definition 5.1 (Strict Algebraic n-Hypergroupoid)**.** A simplicial module $X_\bullet$ is a **strict algebraic n-hypergroupoid** if fillers are unique in dimensions strictly greater than n, and not unique in dimension n. That is:

1. $R_{p,j}(X) = 0$ for all $p > n$ and all j.

2. $R_{n,j}(X) \neq 0$ for at least one j.

Remark 5.2 (Relation to Duskin and Glenn). Condition *(1)* in Definition 5.1 is exactly the n-hypergroupoid condition of Duskin and Glenn: all horns in dimensions $p > n$ admit unique fillers. Condition *(2)* adds a non-uniqueness requirement in dimension n: there exists j with $R_{n,j}(X) \neq 0$. Thus a strict algebraic n-hypergroupoid in our sense is an n-hypergroupoid in which horn fillers are non-unique in dimension n and unique in all dimensions $p > n$.

Theorem 5.3 (Hypergroupoid Classification). *$X_\bullet(\vec{s}; A)$ is a strict algebraic n-hypergroupoid if and only if $k = n$.*

Proof. We use the criterion established in Corollary 3.9: for each p and j, $R_{p,j} \neq 0$ if and only if $k \geq p$.

Condition (1) requires $R_{p,j} = 0$ for all $p > n$. This is equivalent to $k < p$ for all $p \geq n+1$. This holds if and only if $k < n + 1$, i.e., $k \leq n$.

Condition (2) requires $R_{n,j} \neq 0$. This is equivalent to $k \geq n$.

$X_\bullet$ is a strict algebraic n-hypergroupoid if and only if both conditions hold: $(k \leq n) \wedge (k \geq n)$, which is equivalent to $k = n$. □

5.2. Rank Formulas via Inclusion-Exclusion

We analyze the ranks of the components of the DSTM. Since $X_p(\vec{s}; A)$ is a free A-module and the normalization projections are defined over $\mathbb{Z}$ (Remark 4.3), the ranks of the submodules $R_{p,j}, N_p, D_p$ are independent of the ring A.

Let $N_a = n_a - 1$. Recall $n = \min_a N_a$.

Proposition 5.4 (Rank of $R_{p,j}$). *Let $F_j = [p] \setminus \{j\}$. The rank of $R_{p,j}(X)$ is*

$$\operatorname{rank} R_{p,j} = \sum_{t=0}^{p} (-1)^t \binom{p}{t} \prod_{a=1}^{k} (M_a(p) + 1 - t).$$

Proof. By Theorem 3.7 and Proposition 3.8, the rank is the count of $m \in I_p$ such that $\operatorname{im}(m) \supseteq F_j$. We use inclusion–exclusion on $I_p = \prod [M_a(p)]$.

For a subset $S \subseteq F_j$ with $|S| = t$, we count the indices $m \in I_p$ such that $\operatorname{im}(m) \cap S = \emptyset$. Since $M_a(p) \geq p$, $S \subseteq [M_a(p)]$. The number of choices for the a-th component is $|[M_a(p)] \setminus S| = M_a(p) + 1 - t$. The total count of such indices is $\prod_{a=1}^{k} (M_a(p) + 1 - t)$.

Since $|F_j| = p$, the inclusion–exclusion principle yields the formula for the count of indices covering F_j. □

We analyze the normalized complex $(N_\bullet, d_0)$. The cycles are $Z_p(N_\bullet) = \ker(d_0 : N_p \to N_{p-1}) = \bigcap_{i=0}^{p} \ker d_i$.

Corollary 5.5 (Rank of $Z_p(N_\bullet)$). *The rank of the normalized cycles $Z_p(N_\bullet)$ is*

$$\operatorname{rank} Z_p(N_\bullet) = \sum_{t=0}^{p+1} (-1)^t \binom{p+1}{t} \prod_{a=1}^{k} (M_a(p) + 1 - t).$$

Proof. $Z_p(N_\bullet)$ is spanned by basis elements E_m such that $\operatorname{im}(m) \supseteq [p]$. The formula follows by applying inclusion–exclusion with the covered set $[p]$ (size $p + 1$). □

5.3. Constant Shape and Stirling Numbers

If the shape is constant, then $n_a = n + 1$ for all a, where n is the simplicial dimension. In this case $N_a = n$ and hence $M_a(p) = p$ for all a.

The formulas simplify using the Stirling numbers of the second kind $\left\{ {k \atop m} \right\}$ (see, e.g., Stanley [7]).

Theorem 5.6 (Ranks for Constant Shape)**.** *If $M_a(p) = p$ for all a (i.e., $I_p = [p]^k$), the ranks of the components of the Moore complex in degree p are:*

$$\operatorname{rank} Z_p(N_\bullet) = (p+1)! \left\{ {k \atop p+1} \right\},$$
$$\operatorname{rank} N_p(X) = p! \left\{ {k \atop p} \right\} + (p+1)! \left\{ {k \atop p+1} \right\}.$$

Proof. We interpret these ranks in terms of finite differences of the polynomial $P(x) = x^k$.

The rank of $Z_p(N_\bullet)$ specializes to

$$\sum_{t=0}^{p+1} (-1)^t \binom{p+1}{t} (p+1-t)^k = \Delta^{p+1}(x^k)\big|_{x=0}.$$

Using the expansion $x^k = \sum_{m=0}^{k} \left\{ {k \atop m} \right\} x^{\underline{m}}$. The operator acts by $\Delta^{p+1}(x^{\underline{m}}) = m^{\underline{p+1}} x^{\underline{m-p-1}}$. Evaluating at $x = 0$, only the term $m = p + 1$ contributes (since $0^{\underline{0}} = 1$ and $0^{\underline{j}} = 0$ for $j > 0$), yielding $(p+1)! \left\{ {k \atop p+1} \right\}$.

The rank of $N_p(X) = R_{p,0}$ specializes to

$$\sum_{t=0}^{p} (-1)^t \binom{p}{t} (p+1-t)^k = \Delta^{p}(x^k)\big|_{x=1}.$$

Evaluating $\Delta^p(x^{\underline{m}}) = m^{\underline{p}} x^{\underline{m-p}}$ at $x = 1$. The term $1^{\underline{j}}$ is 1 if $j = 0$ or $j = 1$, and 0 if $j \geq 2$. If $m = p$, the contribution is $p! \left\{ {k \atop p} \right\}$. If $m = p + 1$, the contribution is $(p+1)! \left\{ {k \atop p+1} \right\}$. □

5.4. Boundaries and Contractibility

We analyze the boundaries $B_p(N_\bullet) = \operatorname{im}(d_0 : N_{p+1} \to N_p)$.

Proposition 5.7 (Non-triviality of Boundaries)**.** *$B_p(N_\bullet) \neq 0$ if and only if $k \geq p + 1$.*

Proof. If $k < p+1$, then $N_{p+1}(X) = 0$ by Corollary 3.9 (since $N_{p+1} = R_{p+1,0}$), so $B_p(N_\bullet) = 0$.

If $k \geq p + 1$, we construct a non-zero boundary. We seek $T \in N_{p+1}(X)$ such that $d_0(T) \neq 0$. We require an index $m \in I_{p+1}$ such that $E_m \in N_{p+1}(X)$ and $d_0(E_m) \neq 0$. $E_m \in N_{p+1}(X)$ requires $\operatorname{im}(m) \supseteq \{1, \ldots, p+1\}$. $d_0(E_m) \neq 0$ requires $m \in \operatorname{im}(\Delta_0^{p+1})$, which is equivalent to $0 \notin \operatorname{im}(m)$. Thus we seek m such that $\operatorname{im}(m) = \{1, \ldots, p+1\}$. Since $k \geq p+1$, such an m exists. Since $M_a(p+1) \geq p+1$, $m \in I_{p+1}$. Therefore $B_p(N_\bullet) \neq 0$. □

The DSTM $X_\bullet$ is contractible (Theorem Appendix A.4). Consequently, $H_*(X_\bullet) = 0$, implying $Z_p(N_\bullet) = B_p(N_\bullet)$ for all p. The differential $d_0 : N_{p+1} \to N_p$ maps surjectively onto $Z_p(N_\bullet)$ with kernel $Z_{p+1}(N_\bullet)$.

Proposition 5.8 (Rank Consistency Check)**.** *The rank formulas derived satisfy the consistency condition required by contractibility (Rank-Nullity):*

$$\operatorname{Rank} N_{p+1}(X) = \operatorname{Rank} Z_{p+1}(N_\bullet) + \operatorname{Rank} Z_p(N_\bullet).$$

Proof. We verify the identity using the formulas from Theorem 5.6 in the constant shape case.

$$\operatorname{Rank} Z_{p+1} + \operatorname{Rank} Z_p = (p+2)! \left\{ {k \atop p+2} \right\} + (p+1)! \left\{ {k \atop p+1} \right\}.$$

This matches the formula for $\operatorname{Rank} N_{p+1}(X)$ derived in Theorem 5.6 (by replacing p with $p+1$).

For general shape, the identity follows directly from the inclusion–exclusion formulas and the relation $M_a(p+1) = M_a(p) + 1$: writing $A_p(t) := \prod_{a=1}^{k}(M_a(p) + 1 - t)$, Pascal's identity $\binom{p+2}{t} = \binom{p+1}{t} + \binom{p+1}{t-1}$ gives

$$\operatorname{rank} Z_{p+1} = \sum_{t=0}^{p+2} (-1)^t \binom{p+2}{t} A_{p+1}(t) = \sum_{t=0}^{p+1} (-1)^t \binom{p+1}{t} A_{p+1}(t) - \sum_{u=0}^{p+1} (-1)^u \binom{p+1}{u} A_p(u),$$

i.e. $\operatorname{rank} Z_{p+1} = \operatorname{rank} N_{p+1} - \operatorname{rank} Z_p$. □

6. The Horn Non-Degeneracy Lemma and Decomposition

We establish that the horn kernel and the degenerate submodule intersect trivially. This ensures the Horn decomposition is compatible with the standard decomposition of simplicial modules.

Lemma 6.1 (Horn Non-Degeneracy)**.** *For any simplicial module $X_\bullet$, any degree $p \geq 1$, and any $j \in [p]$,*

$$R_{p,j}(X) \cap D_p(X) = \{0\}.$$

Proof. We filter the degenerate submodule $D_p(X)$ by the maximum index of the degeneracy operators. For $0 \leq r \leq p-1$, set

$$H_r := \sum_{i=0}^{r} \operatorname{im}(s_i) \subseteq X_p,$$

and for $0 \leq r \leq p-2$, set

$$H'_r := \sum_{i=0}^{r} \operatorname{im}(s_i) \subseteq X_{p-1},$$

and set $H_{-1} = H'_{-1} = \{0\}$. Note that $D_p(X) = H_{p-1}$.

Let $x \in R_{p,j}(X) \cap D_p(X)$ and assume $x \neq 0$. Choose $m \geq 0$ minimal such that $x \in H_m$. Then we can write

$$x = s_m(z) + y, \qquad z \in X_{p-1}, \quad y \in H_{m-1}.$$

If we can show $z \in H'_{m-1}$, then $z = \sum_{i=0}^{m-1} s_i(u_i)$. Using the identity $s_m s_i = s_i s_{m-1}$ for $i < m$, we obtain

$$s_m(z) = \sum_{i=0}^{m-1} s_m s_i(u_i) = \sum_{i=0}^{m-1} s_i s_{m-1}(u_i) \in H_{m-1}.$$

This implies $x \in H_{m-1}$, contradicting the minimality of m. Thus, it suffices to prove $z \in H'_{m-1}$.

We split the proof into cases based on the index j.

Case 1: $j \neq m+1$. Since $m+1 \leq p$ and $m+1 \neq j$, we have $d_{m+1}(x) = 0$.

$$0 = d_{m+1}(x) = d_{m+1}s_m(z) + d_{m+1}(y) = z + d_{m+1}(y),$$

where we used $d_{m+1}s_m = \mathrm{id}$. Since $y \in H_{m-1}$, we write $y = \sum_{i=0}^{m-1} s_i(w_i)$. For $i < m$, the identity $d_{m+1}s_i = s_i d_m$ holds, so

$$d_{m+1}(y) = \sum_{i=0}^{m-1} s_i d_m(w_i) \in H'_{m-1}.$$

Hence $z = -d_{m+1}(y) \in H'_{m-1}$, as required.

Case 2: $j = m+1$. In this case, the face maps $d_0, \ldots, d_m$ all annihilate x (since $j = m+1$). Since $x \in H_m$, we may write x as a sum $x = \sum_{i=0}^{m} s_i(w_i)$ for some coefficients $w_i \in X_{p-1}$. (Note that z from the setup corresponds to w_m).

We prove by induction on k (for $0 \leq k \leq m$) that $w_k \in H'_{m-1}$.

Base step ($k = 0$): Consider $d_0(x) = 0$.

$$0 = \sum_{i=0}^{m} d_0 s_i(w_i) = d_0 s_0(w_0) + \sum_{i=1}^{m} d_0 s_i(w_i).$$

Using $d_0 s_0 = \mathrm{id}$ and $d_0 s_i = s_{i-1} d_0$ for $i \geq 1$:

$$0 = w_0 + \sum_{i=1}^{m} s_{i-1} d_0(w_i).$$

The sum consists of terms in $\mathrm{im}(s)$, so $w_0 \in H'_{m-1}$.

Inductive step: Assume $w_0, \ldots, w_{k-1} \in H'_{m-1}$ for some $1 \leq k \leq m$. Consider $d_k(x) = 0$.

$$0 = \sum_{i=0}^{m} d_k s_i(w_i).$$

We split the sum into four parts based on the index i:

1. $i < k-1$: $d_k s_i(w_i) = s_i d_{k-1}(w_i) \in H'_{m-1}$.
2. $i = k-1$: $d_k s_{k-1}(w_{k-1}) = w_{k-1}$ (using $d_k s_{k-1} = \mathrm{id}$). By the inductive hypothesis, $w_{k-1} \in H'_{m-1}$.
3. $i = k$: $d_k s_k(w_k) = w_k$ (using $d_k s_k = \mathrm{id}$).
4. $i > k$: $d_k s_i(w_i) = s_{i-1} d_k(w_i) \in H'_{m-1}$.

Substituting these into the equation yields

$$0 = (\text{terms in } H'_{m-1}) + w_{k-1} + w_k + (\text{terms in } H'_{m-1}).$$

Solving for w_k, we see that w_k is a sum of elements in H'_{m-1}. Thus $w_k \in H'_{m-1}$.

By induction, $w_m \in H'_{m-1}$. Since $z = w_m$, we have $z \in H'_{m-1}$.

In all cases, we conclude $z \in H'_{m-1}$, which contradicts the minimality of m unless $x = 0$. □

Theorem 6.2 (Horn decomposition)**.** *For any simplicial module $X_\bullet$, any $p \geq 1$, and any $j \in [p]$, the Moore filler map μ_j (Definition 4.5) is a section of the horn map Φ_j, and*

$$X_p = R_{p,j}(X) \oplus D_p(X).$$

In particular, the image of the filler map is exactly the degenerate submodule: $\mathrm{im}(\mu_j) = D_p(X)$.

Proof. By construction (Definition 4.5) and Proposition 4.6, μ_j is a section of Φ_j, i.e. $\Phi_j \circ \mu_j = \mathrm{id}_{\mathrm{Horns}(p,j)}$. Thus the short exact sequence in Proposition 4.6 splits, and we obtain

$$X_p = R_{p,j}(X) \oplus \mathrm{im}(\mu_j).$$

Definition 4.5 ensures $\mathrm{im}(\mu_j) \subseteq D_p(X)$. Lemma 6.1 ensures that the direct summands intersect trivially: $R_{p,j}(X) \cap D_p(X) = \{0\}$. Now let $d \in D_p(X)$. Decompose it according to the splitting as $d = r + m$ where $r \in R_{p,j}(X)$ and $m \in \mathrm{im}(\mu_j)$. Then $r = d - m$. Since $d \in D_p(X)$ and $m \in \mathrm{im}(\mu_j) \subseteq D_p(X)$, we have $r \in D_p(X)$. Thus $r \in R_{p,j}(X) \cap D_p(X) = \{0\}$, so $r = 0$. Consequently $d = m$, proving that $D_p(X) \subseteq \mathrm{im}(\mu_j)$. Therefore $\mathrm{im}(\mu_j) = D_p(X)$. □

Remark 6.3 (Normalization Conventions). Theorem 6.2 generalizes the classical Normalization Theorem (Theorem 4.2). Standard conventions for the Moore complex (or normalized chain complex) vary:

1. $N_p^{(0)}(X) = \bigcap_{i=0}^{p-1} \ker d_i = R_{p,p}(X)$.
2. $N_p^{(1)}(X) = \bigcap_{i=1}^{p} \ker d_i = R_{p,0}(X)$.

In this paper we adopt the second convention $N_p(X) = N_p^{(1)}(X) = R_{p,0}(X)$. Our horn decomposition encompasses both as the cases $j = p$ and $j = 0$ respectively.

7. Homology Dichotomy and Classification

7.1. The Horn Complex

In this subsection we assume $n \geq 2$; for $n < 2$ the horn complex in low degrees degenerates and the corresponding homology statements are easily checked directly. Fix $j \in [n]$. Define the linear maps (identifying $\bigoplus_{i \neq j} X_{n-1} \cong \prod_{i \neq j} X_{n-1}$):

$$\Phi_j : X_n \longrightarrow \bigoplus_{i \neq j} X_{n-1}, \qquad \Phi_j(T) = (d_i T)_{i \neq j},$$

$$\Psi : \bigoplus_{i \neq j} X_{n-1} \longrightarrow \bigoplus_{\substack{i<m \\ i,m \neq j}} X_{n-2}, \qquad \Psi((x_i)) = (d_i x_m - d_{m-1} x_i)_{i<m}.$$

Proposition 7.1. *The sequence*

$$C_j^{\mathrm{horn}}: \qquad 0 \longrightarrow X_n \xrightarrow{\partial_2=\Phi_j} \bigoplus_{i\neq j} X_{n-1} \xrightarrow{\partial_1=\Psi} \bigoplus_{i<m,\, i,m\neq j} X_{n-2} \longrightarrow 0$$

is a chain complex (indexed homologically with X_n in degree 2). Moreover,

$$H_2\big(C_j^{\mathrm{horn}}\big) \;\cong\; R_{n,j}, \qquad H_1\big(C_j^{\mathrm{horn}}\big) = 0.$$

Proof. $\partial_1 \circ \partial_2 = 0$ by the simplicial identities $d_i d_m = d_{m-1} d_i$ for $i < m$. Since the complex starts in degree 2, $H_2 = \ker \partial_2 = R_{n,j}$. The kernel of ∂_1 is the space of compatible horns $\mathrm{Horns}(n, j)$; by the existence of fillers (Proposition 4.6), $\operatorname{im} \partial_2 = \mathrm{Horns}(n, j)$. Thus $H_1 = \ker \partial_1 / \operatorname{im} \partial_2 = 0$. □

Theorem 7.2 (Short exact sequence in the horn kernel)**.** *The face map $d_j : R_{n,j} \to X_{n-1}$ induces a natural short exact sequence*

$$0 \longrightarrow Z_n(N_\bullet) \longrightarrow R_{n,j} \xrightarrow{d_j} d_j(R_{n,j}) \longrightarrow 0.$$

Proof. The kernel of $d_j|_{R_{n,j}}$ is $R_{n,j} \cap \ker d_j = \bigcap_{i\neq j} \ker d_i \cap \ker d_j = Z_n(N_\bullet)$. □

7.2. The Classification Dichotomy

We use the ranks established in Section 5.

Corollary 7.3 (Filler dichotomy)**.** *The ranks of $R_{n,j}$ and $Z_n(N_\bullet)$ (Proposition 5.4, Corollary 5.5) determine the following classification based on the tensor order k:*

1. *If $k < n$, then $R_{n,j} = 0$ (fillers are unique in dimension n).*
2. *If $k = n$, then $R_{n,j} \neq 0$ and $Z_n(N_\bullet) = 0$. Hence $R_{n,j} \cong d_j(R_{n,j})$.*
3. *If $k \geq n+1$, then $Z_n(N_\bullet) \neq 0$ and $Z_n(N_\bullet)$ injects canonically into $R_{n,j}$.*

Corollary 7.4 (Constant shape ranks)**.** *For the constant shape $\vec{s} = (n+1, \ldots, n+1)$, the ranks are given by Theorem 5.6:*

$$\operatorname{rank} Z_n(N_\bullet) = (n+1)! \left\{ {k \atop n+1} \right\},$$

$$\operatorname{rank} R_{n,j} = n! \left\{ {k \atop n} \right\} + (n+1)! \left\{ {k \atop n+1} \right\}.$$

In particular:

- *$Z_n(N_\bullet) \neq 0$ if and only if $k \geq n+1$.*
- *$R_{n,j} \neq 0$ if and only if $k \geq n$.*

Remark 7.5 (Moore filler vs. T). Write $\Lambda_j^n(T) := \Phi_j(T) \in \mathrm{Horns}(n, j)$ for the (n, j)-horn of T. Moore's filler T' satisfies $T' = T$ if and only if $T \in D_n$: the "only if" direction holds because T' is always degenerate (Definition 4.5), and the "if" direction follows from Lemma 6.1, since in that case $T - T'$ lies in $R_{n,j} \cap D_n = \{0\}$. If $k < n$, then $R_{n,j} = 0$, so $T' = T$ always holds (and $D_n = X_n$). If $k \geq n$ and $T \notin D_n$, then $T - T' \in R_{n,j} \setminus \{0\}$ and is supported on the missing indices (Theorem 3.7).

7.3. Interpretation as Algebraic n-Hypergroupoids

We adapt the definition of Duskin (1979) and Glenn (1982).

Definition 7.6 (Algebraic n-hypergroupoid)**.** A simplicial module $X_\bullet$ is an **algebraic n-hypergroupoid** if horn fillers are unique in dimensions $p > n$. It is **strict** if it is not an $(n-1)$-hypergroupoid (i.e., fillers in dimension n are not unique).

Proposition 7.7. *A simplicial module $X_\bullet$ is an algebraic n-hypergroupoid if and only if $R_{p,j}(X) = 0$ for all $p > n$ and all j.*

Proof. Simplicial modules are Kan complexes (fillers always exist, e.g., Proposition 4.6). As observed in Section 3, the set of fillers of a given horn is an affine torsor modeled on $R_{p,j}$, hence a singleton if and only if $R_{p,j} = 0$. □

Applying the criterion of Corollary 3.9 to the DSTM $X_\bullet(\vec{s}; A)$ with simplicial dimension n yields the hypergroupoid classification of Theorem 5.3: $X_\bullet(\vec{s}; A)$ is a strict algebraic n-hypergroupoid if and only if the tensor order k equals n.

7.4. The Subcomplex of Normalized Cycles

We analyze the subcomplex formed by the normalized cycles $Z_r(N_\bullet)$. In the DSTM, $Z_r(N_\bullet)$ is spanned by basis elements E_m such that $\mathrm{im}(m) \supseteq [r]$.

Lemma 7.8. *The family $(Z_r(N_\bullet))_{r\geq 0}$ forms a chain subcomplex of $(X_\bullet, \partial)$. Furthermore, the differential on this subcomplex is zero:*

$$\partial_r\big(Z_r(N_\bullet)\big) = 0, \qquad H_r\big(Z_\bullet(N_\bullet)\big) \cong Z_r(N_\bullet).$$

There is a short exact sequence of chain complexes

$$0 \longrightarrow Z_\bullet(N_\bullet) \longrightarrow X_\bullet \longrightarrow X_\bullet/Z_\bullet(N_\bullet) \longrightarrow 0. \tag{1}$$

Proof. By definition, $Z_r(N_\bullet) = \bigcap_{i=0}^{r} \ker d_i$. If $x \in Z_r(N_\bullet)$, then $d_i(x) = 0$ for all i. Therefore the boundary $\partial_r(x) = \sum (-1)^i d_i(x) = 0$. This confirms that $Z_\bullet(N_\bullet)$ is a chain subcomplex with zero differential. The homology of a complex with zero differential is the complex itself. □

Corollary 7.9 (Quotient homology)**.** *For the DSTM $X_\bullet(\vec{s}; A)$, which is contractible by Theorem Appendix A.4, the short exact sequence of chain complexes (1) induces isomorphisms*

$$H_r\big(X_\bullet/Z_\bullet(N_\bullet)\big) \;\cong\; H_{r-1}\big(Z_\bullet(N_\bullet)\big) \cong Z_{r-1}(N_\bullet)$$

for all $r \geq 1$.

Proof. By Theorem Appendix A.4, the DSTM $X_\bullet(\vec{s}; A)$ is contractible, hence $H_r(X_\bullet) = 0$ for all $r \geq 0$. Taking homology of the short exact sequence (1) gives a long exact sequence

$$\cdots \to H_r(Z_\bullet(N_\bullet)) \to H_r(X_\bullet) \to H_r(X_\bullet/Z_\bullet(N_\bullet)) \to H_{r-1}(Z_\bullet(N_\bullet)) \to \cdots.$$

The differential on $Z_\bullet(N_\bullet)$ is zero, so $H_r(Z_\bullet(N_\bullet)) \cong Z_r(N_\bullet)$ for all r. Using $H_r(X_\bullet) = 0$ and exactness, we obtain

$$H_r\big(X_\bullet/Z_\bullet(N_\bullet)\big) \;\cong\; H_{r-1}\big(Z_\bullet(N_\bullet)\big) \cong Z_{r-1}(N_\bullet).$$

□

Remark 7.10 (Homology spheres). For a normalized cycle $C \in Z_n(N_\bullet)$, the generated simplicial subobject $\langle C \rangle \subseteq X_\bullet$ (the degeneracy-closure) provides an algebraic model of the n-sphere. If C is primitive (e.g., a basis element E_m over a PID A), $\langle C \rangle$ is isomorphic to the simplicial sphere $A[\mathbf{\Delta}^n]/\operatorname{Sk}_{n-1}(A[\mathbf{\Delta}^n])$. It satisfies $H_n(\langle C \rangle) \cong A$ and $H_m(\langle C \rangle) = 0$ for $m \neq n$. (See Weibel [4, Exercise 8.3.4].)

8. The Geometry of Generated Subobjects and Moduli Spaces

We analyze the isomorphism classes of the simplicial submodules $\langle T \rangle \subseteq X_\bullet(\vec{s}; K)$ generated by a tensor $T \in X_n$. We assume the ground ring K is an infinite field.

8.1. Realization Maps and Kernel Sequences

Let $C_\bullet := K[\mathbf{\Delta}^n]$ be the standard simplicial K-module representing the n-simplex. In degree p, $C_p = K[\mathbf{\Delta}^n]_p$ is the free K-vector space with basis the set of morphisms $\mathbf{\Delta}([p],[n])$. The dimension is $S_p := \dim(C_p) = \binom{n+p+1}{p+1}$. Let $\iota_n \in C_n$ be the generator corresponding to the identity map.

Definition 8.1 (Realization Map and Kernel Sequence)**.** For $T \in X_n(\vec{s}; K)$, the **realization map** $f_T : C_\bullet \to X_\bullet(\vec{s}; K)$ is the unique morphism of simplicial modules sending ι_n to T. The **kernel sequence** of T is the simplicial submodule

$$K(T)_\bullet := \ker(f_T) \subset C_\bullet,$$

defined degreewise by $K(T)_p := \ker(f_{T,p}) \subseteq C_p$.

The image is $\langle T \rangle_\bullet = \operatorname{im}(f_T)$, and the realization map induces an isomorphism of simplicial K-modules

$$C_\bullet / K(T)_\bullet \xrightarrow{\cong} \langle T \rangle_\bullet.$$

The isomorphism class of $\langle T \rangle_\bullet$ is therefore determined by its kernel sequence $K(T)_\bullet$.

Remark 8.2 (Finite Generation). Since $K[\mathbf{\Delta}^n]$ is generated in degree n, any simplicial submodule is determined by its components up to degree n.

Proposition 8.3 (Quotients and generated submodules)**.** *For each $T \in X_n(\vec{s}; K)$, the realization map $f_T : C_\bullet \to X_\bullet(\vec{s}; K)$ induces an isomorphism*

$$C_\bullet / K(T)_\bullet \xrightarrow{\cong} \langle T \rangle_\bullet.$$

In particular, all invariants of the generated submodule $\langle T \rangle_\bullet$ can be computed from its kernel sequence $K(T)_\bullet \subseteq C_\bullet$.

Proof. Since $K(T)_\bullet = \ker(f_T)$, the universal property of quotients gives a simplicial map

$$C_\bullet / K(T)_\bullet \longrightarrow \langle T \rangle_\bullet$$

which is an isomorphism in each degree, because $K(T)_p = \ker(f_{T,p})$ and $\langle T \rangle_p = \operatorname{im}(f_{T,p})$. □

Remark 8.4. For our purposes it suffices that each generated submodule $\langle T \rangle_\bullet$ arises as a quotient $C_\bullet / K(T)_\bullet$, and that all invariants we study (e.g. homology groups and the incidence conditions defining the moduli space) depend only on the kernel sequence $K(T)_\bullet \subseteq C_\bullet$. We make no assertion that isomorphic quotients have isomorphic kernels, nor that every isomorphism between quotients lifts to an automorphism of $C_\bullet$.

8.2. Index Collisions and the Realization Matrix

Let $R_p := \dim(X_p)$.

Definition 8.5 (Index collision map)**.** For each $p \geq 0$, the **index collision map** is

$$\mathcal{I}_p : I_p \times \mathbf{\Delta}([p],[n]) \longrightarrow I_n, \qquad \mathcal{I}_p(m,\alpha) := I_\bullet(\alpha)(m).$$

Definition 8.6 (Realization matrix)**.** The **realization matrix** $M_{T,p} \in \mathrm{Mat}_{R_p \times S_p}(K)$ represents $f_{T,p}$. Its entry in row $m \in I_p$ and column $\alpha \in \mathbf{\Delta}([p],[n])$ is

$$M_{T,p}[m,\alpha] = (f_{T,p}(\alpha))(m) = T_{\mathcal{I}_p(m,\alpha)}.$$

A **collision** in degree p is a pair $(m,\alpha) \neq (m',\alpha')$ with $\mathcal{I}_p(m,\alpha) = \mathcal{I}_p(m',\alpha')$. Collisions force the corresponding entries of $M_{T,p}$ to agree for every T, and are the source of rank loss in the symbolic matrices $M_{T_{\mathrm{sym}},p}$.

8.3. The Moduli Map and Grassmannians

Let $\mathcal{V} = \{v_u\}_{u \in I_n}$ be indeterminates and $\mathbb{K} := K(\mathcal{V})$ the function field. The symbolic tensor T_{sym} has entries v_u.

Definition 8.7 (Generic Rank and Kernel Dimension)**.** The **generic rank** $R'_p(\vec{s})$ is the rank of the symbolic matrix $M_{T_{\mathrm{sym}},p}$ over $\mathbb{K}$. The **generic kernel dimension** is $K'_p(\vec{s}) := S_p - R'_p(\vec{s})$.

Definition 8.8 (Moduli Map)**.** The **generic locus** $\mathcal{U} \subset X_n(\vec{s};K)$ is the (non-empty) Zariski open set where $\mathrm{rank}(M_{T,p}) = R'_p(\vec{s})$ for all $0 \leq p \leq n$.

The **Grassmannian associated to the shape** $\vec{s}$ is

$$\mathrm{Gr}(\vec{s}) := \prod_{\substack{0 \leq p \leq n \\ K'_p(\vec{s})>0}} \mathrm{Gr}\big(K'_p(\vec{s}), K[\mathbf{\Delta}^n]_p\big).$$

By Remark 8.2, a simplicial submodule of $C_\bullet = K[\mathbf{\Delta}^n]$ is determined by its components in degrees $0 \leq p \leq n$, so recording the kernels $K(T)_p = \ker f_{T,p}$ in this range determines the entire kernel sequence $K(T)_\bullet$.

The **Moduli Map** Ψ is the algebraic map:

$$\Psi : \mathcal{U} \longrightarrow \mathrm{Gr}(\vec{s}), \quad T \mapsto (\ker f_{T,p})_{\substack{0 \leq p \leq n \\ K'_p(\vec{s})>0}}.$$

The **Moduli Space of Kernel Sequences** $\mathcal{M}(\vec{s})$ is the image $\Psi(\mathcal{U})$.

8.4. Injectivity Analysis and Examples

Lemma 8.9. *The functor $I_\bullet : \mathbf{\Delta} \to \mathbf{Set}$ preserves monomorphisms.*

Proof. Monomorphisms in $\mathbf{\Delta}$ are generated by coface maps δ_i. $I_\bullet(\delta_i^p) = \Delta_i^p$ is a product of injections, hence injective. □

Theorem 8.10 (Injectivity and Dominance at $p = 0$)**.** *In the DSTM:*

1. *The map $\mathcal{I}_0$ is injective.* $R'_0 = \min(R_0, S_0)$.
2. *If $K'_0 > 0$, the projection $\Psi_0 : \mathcal{U} \to \mathrm{Gr}(K'_0, K[\mathbf{\Delta}^n]_0)$ is dominant.*

Proof. (1) $S_0 = n+1$. Let a_0 be an axis where $n_{a_0} = n+1$. Then $M_{a_0}(0) = 0$. For $m \in I_0$, $m_{a_0} = 0$. Let $\alpha_i : [0] \to [n]$ be the vertex map $\alpha_i(0) = i$. The a_0-th coordinate of $\mathcal{I}_0(m, \alpha_i)$ is $\alpha_i(m_{a_0}) = i$. If $\mathcal{I}_0(m, \alpha_i) = \mathcal{I}_0(m', \alpha_{i'})$, then $i = i'$. By Lemma 8.9, $I_\bullet(\alpha_i)$ is injective, so $m = m'$.

(2) Since $\mathcal{I}_0$ is injective, the entries of $M_{T_{\mathrm{sym}},0}$ are distinct variables. The map $L_0 : X_n \to \mathrm{Mat}_{R_0 \times S_0}(K)$ sending T to $M_{T,0}$ is a projection onto these coordinates, hence surjective. The map from the space of matrices of maximal rank R'_0 to the Grassmannian of their kernels is dominant. Ψ_0 is the composition of L_0 (restricted to $\mathcal{U}$) and this dominant map. □

Lemma 8.11 (Non-injectivity for $p \geq 1$)**.** *For $p \geq 1$, $\mathcal{I}_p$ is not injective.*

Proof. Let $\alpha_0 : [p] \to [n]$ be the constant map to 0. It factors as $\alpha_0 = \iota_0 \circ \pi_0$. Since $p \geq 1$, $|I_p| > |I_0|$. The map $I_\bullet(\pi_0) : I_p \to I_0$ cannot be injective. There exist $m \neq m'$ such that $I_\bullet(\pi_0)(m) = I_\bullet(\pi_0)(m')$. Applying $I_\bullet(\iota_0)$ yields $\mathcal{I}_p(m, \alpha_0) = \mathcal{I}_p(m', \alpha_0)$. □

Example 8.12 (Rank Drop: Shape (2,2))*.* $n = 1, k = 2$. $S = (2, 3)$. $R = (1, 4)$. $p = 1$. $S_1 = 3, R_1 = 4$. The morphisms $\mathbf{\Delta}([1], [1])$ are $\{\mathrm{id}, (0, 0), (1, 1)\}$. $I_1 = [1]^2$. The symbolic realization matrix $M_{T_{\mathrm{sym}},1}$ is:

$$M = \begin{pmatrix} v_{00} & v_{00} & v_{11} \\ v_{01} & v_{00} & v_{11} \\ v_{10} & v_{00} & v_{11} \\ v_{11} & v_{00} & v_{11} \end{pmatrix}.$$

All four 3×3 minors vanish identically over $\mathbb{K}$. The generic rank is $R'_1 = 2$. $K'_1 = 1$.

8.5. The Incidence Variety and Segre–Plücker Embedding

The image $\mathcal{M}(\vec{s})$ lies within the closed incidence subvariety $\mathbf{Gr}^{\mathrm{simp}}(\vec{s}) \subset \mathrm{Gr}(\vec{s})$ defined by the simplicial compatibility conditions.

Theorem 8.13 (Global Structure of the Moduli Space)**.** *$\mathcal{M}(\vec{s})$ is a constructible set contained in $\mathbf{Gr}^{\mathrm{simp}}(\vec{s})$. The projection $\mathcal{M}(\vec{s}) \to \mathrm{Gr}(K'_0(\vec{s}), S_0)$ is dominant (if $K'_0(\vec{s}) > 0$). The moduli space $\mathcal{M}(\vec{s})$ is irreducible and unirational.*

Proof. $\mathcal{U}$ is an open subset of an affine space, hence irreducible. $\mathcal{M}(\vec{s})$ is the image of an irreducible variety under a rational map, hence it is constructible (Chevalley's theorem), irreducible, and unirational. Dominance at $p = 0$ follows from Theorem 8.10. □

Proposition 8.14 (Segre–Plücker linearity)**.** *Embed $\mathrm{Gr}(\vec{s})$ into a projective space via the composition of the Plücker embeddings $\iota_p : \mathrm{Gr}(K'_p, S_p) \hookrightarrow \mathbb{P}(\Lambda^{K'_p} K[\mathbf{\Delta}^n]_p)$ and the Segre embedding Σ. The image $\Sigma(\mathbf{Gr}^{\mathrm{simp}}(\vec{s}))$ is the intersection of $\Sigma(\mathrm{Gr}(\vec{s}))$ with a linear subspace of the ambient projective space.*

Proof. We analyze the condition $d_i(L_p) \subseteq L_{p-1}$. Let $C_q = K[\mathbf{\Delta}^n]_q$. Let ω_p and ω_{p-1} be the Plücker coordinates representing L_p and L_{p-1}. The condition is equivalent to $\Lambda^{K'_p} d_i(\omega_p) \wedge \eta_{p-1} = 0$, where η_{p-1} is a volume element of a complement to L_{p-1}. Under the Segre embedding, these bilinear equations in the Plücker coordinates become linear equations in the homogeneous coordinates of the tensor product space. □

Proposition 8.15 (Determinantal structure of the incidence variety)**.** *Let $C_p := K[\mathbf{\Delta}^n]_p$. On $\mathrm{Gr}(\vec{s})$, let $\mathcal{S}_p \subset C_p \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})}$ denote the pullback of the tautological rank-$K'_p(\vec{s})$ subbundle on $\mathrm{Gr}(K'_p(\vec{s}), C_p)$ along the projection $\mathrm{Gr}(\vec{s}) \to \mathrm{Gr}(K'_p(\vec{s}), C_p)$, and let $\mathcal{S}_{p-1}$ be defined similarly in degree $p-1$.*

For each pair (p,i) with $K'_p(\vec{s}) > 0$, the face map $d_i : C_p \to C_{p-1}$ induces a morphism of vector bundles

$$\varphi_{p,i} : \mathcal{S}_p \longrightarrow \big(C_{p-1} \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})}\big)/\,\mathcal{S}_{p-1}.$$

Then the incidence variety $\mathbf{Gr}^{\mathrm{simp}}(\vec{s})$ is the scheme-theoretic intersection

$$\mathbf{Gr}^{\mathrm{simp}}(\vec{s}) = \bigcap_{p,i} \{\, \varphi_{p,i} = 0 \,\},$$

i.e. it is the intersection of the rank-0 determinantal loci of the bundle maps $\varphi_{p,i}$. In particular, $\mathbf{Gr}^{\mathrm{simp}}(\vec{s})$ is a determinantal subvariety of $\mathrm{Gr}(\vec{s})$.

Proof. A point of $\mathrm{Gr}(\vec{s})$ is a tuple of subspaces $(L_p)_p$ with $L_p \subset C_p$ and $\dim L_p = K'_p(\vec{s})$ whenever $K'_p(\vec{s}) > 0$. By construction of the tautological subbundle, the fiber $(\mathcal{S}_p)_x$ over a point x corresponding to $(L_p)_p$ is exactly L_p, and similarly $(\mathcal{S}_{p-1})_x = L_{p-1}$.

The fixed linear map $d_i : C_p \to C_{p-1}$ induces a morphism of vector bundles

$$d_i \otimes \mathrm{id} : C_p \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})} \longrightarrow C_{p-1} \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})}.$$

Composing with the quotient map $C_{p-1} \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})} \twoheadrightarrow (C_{p-1} \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})})/\mathcal{S}_{p-1}$ and restricting to $\mathcal{S}_p \subset C_p \otimes \mathcal{O}$ gives a bundle map

$$\varphi_{p,i} : \mathcal{S}_p \to (C_{p-1} \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})})/\mathcal{S}_{p-1}.$$

At a point $x = (L_p)_p$, the fiber map $(\varphi_{p,i})_x : L_p \to C_{p-1}/L_{p-1}$ is the composition

$$L_p \hookrightarrow C_p \xrightarrow{d_i} C_{p-1} \twoheadrightarrow C_{p-1}/L_{p-1}.$$

Thus $(\varphi_{p,i})_x = 0$ if and only if $d_i(L_p) \subseteq L_{p-1}$. By definition, $\mathbf{Gr}^{\mathrm{simp}}(\vec{s})$ is the locus of tuples satisfying these inclusions for all (p,i).

In local trivializations, $\varphi_{p,i}$ is represented by a matrix of regular functions, and the condition $(\varphi_{p,i})_x = 0$ corresponds to the vanishing of all entries (all 1×1 minors). Globally, this condition defines the zero locus of the section $\varphi_{p,i}$ of the bundle $\mathcal{H}om(\mathcal{S}_p, \mathcal{Q}_{p-1})$, where we define the quotient bundle $\mathcal{Q}_{p-1} := (C_{p-1} \otimes \mathcal{O}_{\mathrm{Gr}(\vec{s})})/\mathcal{S}_{p-1}$. In terms of homogeneous coordinates, this condition is bilinear in the Plücker coordinates of the factors (and thus linear in the Segre coordinates), confirming that $\mathbf{Gr}^{\mathrm{simp}}(\vec{s})$ is a determinantal subvariety. □

8.6. Homology of Generated Subobjects

Fix $T \in X_n(\vec{s}; K)$ and write $C_\bullet := K[\mathbf{\Delta}^n]$ and $K_\bullet := K(T)_\bullet = \ker(f_T) \subseteq C_\bullet$. We analyze $H_\bullet(\langle T \rangle_\bullet)$ using the short exact sequence

$$0 \longrightarrow K_\bullet \longrightarrow C_\bullet \longrightarrow \langle T \rangle_\bullet \longrightarrow 0.$$

Recall that $H_p(C_\bullet) = 0$ for $p > 0$ and $H_0(C_\bullet) \cong K$.

Proposition 8.16 (Homology of the generated subobject)**.** *Let $B_p(C_\bullet)$ and $B_p(K_\bullet)$ denote the boundary subspaces in $C_\bullet$ and $K_\bullet$, respectively. Then:*

1. *For $p \geq 2$, there is a natural isomorphism*
$$H_p(\langle T \rangle_\bullet) \cong H_{p-1}(K_\bullet).$$
2. $H_1(\langle T \rangle_\bullet) \cong \big(K_0 \cap B_0(C_\bullet)\big)/B_0(K_\bullet)$.
3. $H_0(\langle T \rangle_\bullet) \cong C_0/\big(K_0 + B_0(C_\bullet)\big)$.

Proof. Applying homology to
$$0 \to K_\bullet \to C_\bullet \to \langle T \rangle_\bullet \to 0$$
gives a long exact sequence. For $p \geq 2$ we have $H_p(C_\bullet) = H_{p-1}(C_\bullet) = 0$, so the connecting homomorphism induces an isomorphism $H_p(\langle T \rangle_\bullet) \cong H_{p-1}(K_\bullet)$, which proves (1).

For $p = 1$ and $p = 0$, the long exact sequence reads
$$0 \to H_1(\langle T \rangle_\bullet) \to H_0(K_\bullet) \to H_0(C_\bullet) \to H_0(\langle T \rangle_\bullet) \to 0,$$
and the standard identification of H_0 as cycles modulo boundaries in degree 0 yields the descriptions in (2) and (3). □

Corollary 8.17 (Generic connectivity)**.** *If $K_0'(\vec{s}) > 0$, then for a generic tensor $T \in \mathcal{U}$, $H_0(\langle T \rangle_\bullet) = 0$.*

Proof. $B_0(C_\bullet)$ is a hyperplane in C_0. By Proposition 8.16, we have $H_0(\langle T \rangle_\bullet) \neq 0$ if and only if $K_0(T) \subseteq B_0(C_\bullet)$. This condition cuts out a proper closed subvariety of $\mathrm{Gr}(K_0'(\vec{s}), S_0)$. Since Ψ_0 is dominant (Theorem 8.10), the generic kernel $K_0(T)$ avoids this locus. □

8.7. Example: Shape $(3, 3)$

Let $\vec{s} = (3, 3)$, so $k = 2$ and $n = 2$. Then X_2 consists of 3×3 matrices. We write $T = (v_{ij})_{0 \leq i,j \leq 2}$. The dimensions are
$$S = (3, 6, 10), \qquad R = (1, 4, 9).$$

We fix the following bases:

- For X_2, the basis $I_2 = [2]^2$ in lexicographic order:
$$(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2).$$
- For $K[\mathbf{\Delta}^2]_1$, the basis of monotone maps $[1] \to [2]$ ordered as
$$(0,0),\ (0,1),\ (0,2),\ (1,1),\ (1,2),\ (2,2).$$
- For $K[\mathbf{\Delta}^2]_2$, the basis of monotone maps $[2] \to [2]$ ordered as the ten non-decreasing triples (a_0, a_1, a_2) with $0 \leq a_0 \leq a_1 \leq a_2 \leq 2$:
$$(0,0,0), (0,0,1), (0,0,2), (0,1,1), (0,1,2), (0,2,2), (1,1,1), (1,1,2), (1,2,2), (2,2,2).$$

Degree $p = 0$.. Here $R_0 = 1$ and $S_0 = 3$. The realization matrix is

$$M_{T,0} = \begin{pmatrix} v_{00} & v_{11} & v_{22} \end{pmatrix}.$$

For generic T, its rank is 1, so $K_0' = 2$. The kernel is the 2-dimensional subspace of K^3 of solutions to

$$v_{00}x_0 + v_{11}x_1 + v_{22}x_2 = 0.$$

On the open chart $v_{00} \neq 0$, it is spanned by

$$(-v_{11}/v_{00},\, 1,\, 0), \qquad (-v_{22}/v_{00},\, 0,\, 1).$$

Thus $K_0(T)$ depends only on the projective class $[v_{00} : v_{11} : v_{22}] \in \mathbb{P}^2$.

Degree $p = 1$.. Here $R_1 = 4$ and $S_1 = 6$. With the bases above, the realization matrix is

$$M_{T,1} = \begin{pmatrix} v_{00} & v_{00} & v_{00} & v_{11} & v_{11} & v_{22} \\ v_{00} & v_{01} & v_{02} & v_{11} & v_{12} & v_{22} \\ v_{00} & v_{10} & v_{20} & v_{11} & v_{21} & v_{22} \\ v_{00} & v_{11} & v_{22} & v_{11} & v_{22} & v_{22} \end{pmatrix}.$$

Two kernel vectors can be written in terms of the diagonal entries. Set

$$c_1^{(1)} := \big(-v_{11}/v_{00},\, 0,\, 0,\, 1,\, 0,\, 0\big)^T, \qquad c_2^{(1)} := \big(-v_{22}/v_{00},\, 0,\, 0,\, 0,\, 0,\, 1\big)^T.$$

A direct calculation shows

$$M_{T,1}\, c_1^{(1)} = 0, \qquad M_{T,1}\, c_2^{(1)} = 0$$

as identities in the polynomial ring $K[v_{ij}]$ (on the open set $v_{00} \neq 0$). Hence $K_1' \geq 2$.

On the other hand, taking e.g.

$$T = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix},$$

one obtains

$$M_{T,1} = \begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 2 & 2 & 2 & 3 \\ 1 & 2 & 3 & 2 & 3 & 3 \end{pmatrix},$$

which has rank 4 (one checks that some 4×4 minor is non-zero). Thus the generic rank is $R_1' = 4$, and therefore $K_1' = S_1 - R_1' = 2$.

In particular, for generic T, the kernel $K_1(T) = \ker f_{T,1}$ is spanned by $c_1^{(1)}$ and $c_2^{(1)}$, so it depends only on (v_{00}, v_{11}, v_{22}).

Degree $p = 2$.. Here $R_2 = 9$ and $S_2 = 10$. With the bases chosen above, the symbolic realization matrix $M_{T,2}$ is

$$M_{T,2} = \begin{pmatrix} v_{00} & v_{00} & v_{00} & v_{00} & v_{00} & v_{00} & v_{11} & v_{11} & v_{11} & v_{22} \\ v_{00} & v_{00} & v_{00} & v_{01} & v_{01} & v_{02} & v_{11} & v_{11} & v_{12} & v_{22} \\ v_{00} & v_{01} & v_{02} & v_{01} & v_{02} & v_{02} & v_{11} & v_{12} & v_{12} & v_{22} \\ v_{00} & v_{00} & v_{00} & v_{10} & v_{10} & v_{20} & v_{11} & v_{11} & v_{21} & v_{22} \\ v_{00} & v_{00} & v_{00} & v_{11} & v_{11} & v_{22} & v_{11} & v_{11} & v_{22} & v_{22} \\ v_{00} & v_{01} & v_{02} & v_{11} & v_{12} & v_{22} & v_{11} & v_{12} & v_{22} & v_{22} \\ v_{00} & v_{10} & v_{20} & v_{10} & v_{20} & v_{20} & v_{11} & v_{21} & v_{21} & v_{22} \\ v_{00} & v_{10} & v_{20} & v_{11} & v_{21} & v_{22} & v_{11} & v_{21} & v_{22} & v_{22} \\ v_{00} & v_{11} & v_{22} & v_{11} & v_{22} & v_{22} & v_{11} & v_{22} & v_{22} & v_{22} \end{pmatrix}.$$

Again there are two explicit kernel vectors depending only on the diagonal. Set

$$c_1^{(2)} := \big(-v_{11}/v_{00},\, 0,\, 0,\, 0,\, 0,\, 0,\, 1,\, 0,\, 0,\, 0\big)^T, \qquad c_2^{(2)} := \big(-v_{22}/v_{00},\, 0,\, 0,\, 0,\, 0,\, 0,\, 0,\, 0,\, 0,\, 1\big)^T.$$

A direct substitution into the matrix above shows

$$M_{T,2}\, c_1^{(2)} = 0, \qquad M_{T,2}\, c_2^{(2)} = 0$$

as polynomial identities (again on the open chart $v_{00} \neq 0$). Thus $K_2' \geq 2$ and $R_2' \leq 10-2 = 8$.

To see that the generic rank is exactly 8, we exhibit a single tensor for which $M_{T,2}$ has rank 8. Take for example

$$T = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}.$$

Substituting these values into $M_{T,2}$ gives a 9×10 numerical matrix with rank 8. Since rank is upper semicontinuous in families, this shows that the generic rank is $R_2' = 8$, and therefore $K_2' = S_2 - R_2' = 2$.

Moreover, the two kernel generators $c_1^{(2)}, c_2^{(2)}$ depend only on v_{00}, v_{11}, v_{22}, not on the off-diagonal entries.

Collecting the three degrees, we have:

Proposition 8.18 (Generic kernel dimensions for shape $(3,3)$)**.** *For $\vec{s} = (3,3)$, the generic kernel dimensions are*

$$K'(\vec{s}) = (K_0', K_1', K_2') = (2,2,2).$$

On the open chart $v_{00} \neq 0$, the kernels $K_p(T)$ are generated by vectors whose coordinates are rational functions of v_{11}/v_{00} and v_{22}/v_{00} only.

Corollary 8.19 (Moduli space for shape $(3,3)$)**.** *For $\vec{s} = (3,3)$, the moduli space $\mathcal{M}(\vec{s})$ is a surface of dimension 2, birational to $\mathbb{P}^2$.*

Proof. The generic kernel vector in degree 0 is the hyperplane

$$K_0(T) = \{(x_0, x_1, x_2) \mid v_{00}x_0 + v_{11}x_1 + v_{22}x_2 = 0\} \subset K^3.$$

Thus the point $K_0(T) \in \mathrm{Gr}(2,3)$ determines the projective class $[v_{00} : v_{11} : v_{22}] \in \mathbb{P}^2$, and hence determines the ratios v_{11}/v_{00} and v_{22}/v_{00} on the chart $v_{00} \neq 0$. By the explicit formulas above, $K_1(T)$ and $K_2(T)$ are then uniquely determined. Therefore, on this chart, the moduli map Ψ factors through the projection

$$X_2 \dashrightarrow \mathbb{P}^2, \qquad T \longmapsto [v_{00} : v_{11} : v_{22}],$$

and the induced map $\mathbb{P}^2 \dashrightarrow \mathcal{M}(\vec{s})$ is birational onto its image. Hence $\mathcal{M}(\vec{s})$ has dimension 2 and is birational to $\mathbb{P}^2$. □

9. Rank shifts and homology of submodules

Let K be an arbitrary field. A rank shift enlarges the tensor axes while retaining the simplicial degree and its face and degeneracy indices. The resulting modules contain explicit representations of graph chain complexes. Their tensor products also give representations of reduced smash products. Homology of a simplicial module always means the homology of its normalized chain complex.

9.1. Fixed tails and normalized chains

Fix $k \geq 1$ and a tuple $\mathbf{c} = (c_1, \ldots, c_k)$ of nonnegative integers. For $p \geq 0$, put

$$I_p(\mathbf{c}) = \prod_{a=1}^{k} [p + c_a], \qquad X_p(\mathbf{c}) = K^{I_p(\mathbf{c})}.$$

The coordinates $0, \ldots, p$ on each axis form its simplicial part; the remaining c_a coordinates form a fixed tail. For an order-preserving map $\alpha : [q] \to [p]$, define

$$\alpha^{+c_a} : [q + c_a] \longrightarrow [p + c_a], \qquad \alpha^{+c_a}(j) = \begin{cases} \alpha(j), & 0 \leq j \leq q, \\ p + t, & j = q + t, \quad 1 \leq t \leq c_a. \end{cases}$$

This is the ordinal sum of α with the identity on a c_a-element ordered set. Define

$$(\alpha^* T)(j_1, \ldots, j_k) = T\big(\alpha^{+c_1}(j_1), \ldots, \alpha^{+c_k}(j_k)\big). \tag{2}$$

Since $(\alpha\beta)^{+c_a} = \alpha^{+c_a}\beta^{+c_a}$, these maps make $X(\mathbf{c})$ a simplicial K-module. For $p \geq 1$ its degree-p faces are $d_0, \ldots, d_p$, and for $p \geq 0$ its degree-p degeneracies are $s_0, \ldots, s_p$. Thus d_i deletes coordinate i on every axis and s_i duplicates coordinate i on every axis. The tail lengths do not increase the number of face or degeneracy operators.

Replacing $\mathbf{c}$ by $\mathbf{c} + r\mathbf{1}$, where $r \geq 0$ and $\mathbf{1} = (1, \ldots, 1)$, is a common rank shift. We write

$$X^{(k,r)} = X(r, \ldots, r), \qquad X_p^{(k,r)} = K^{[p+r]^k}$$

for the equal-tail case. These definitions also include unequal axis lengths without a common shift. The DSTM of the preceding sections is the case $\min_a c_a = 0$. Allowing all tail lengths to be positive extends that construction by enlarging its degree-zero module.

For $m = (m_1, \dots, m_k) \in I_p(\mathbf{c})$, let E_m denote the tensor unit at m, and put $\mathrm{val}(m) = \{m_1, \dots, m_k\}$. We use the Moore normalization

$$N_pX = \bigcap_{i=1}^{p} \ker d_i, \qquad \partial = d_0 : N_pX \longrightarrow N_{p-1}X \quad (p \geq 1).$$

Here $N_0X = X_0$ and the complex is unaugmented.

Proposition 9.1 (Normalized basis and ambient homology)**.** *For $p \geq 1$, the following sets are bases of the indicated spaces:*

$$\begin{aligned} N_pX(\mathbf{c}): &\quad \{E_m : \{1, \dots, p\} \subseteq \mathrm{val}(m)\}, \\ Z_p(NX(\mathbf{c})): &\quad \{E_m : \{0, \dots, p\} \subseteq \mathrm{val}(m)\}. \end{aligned}$$

Moreover,

$$H_p(NX(\mathbf{c})) = 0 \quad (p > 0), \qquad H_0(NX(\mathbf{c})) \cong K^{\prod_{a=1}^{k} \{1, \dots, c_a\}}. \tag{3}$$

In particular, $\dim_K H_0 = \prod_{a=1}^{k} c_a$*, with value zero if any* $c_a = 0$*.*

Proof. The image d_iE_m is zero precisely when some coordinate of m equals i. On the tensor units whose coordinates all avoid i, the map d_i is a bijection onto the tensor-unit basis in degree $p-1$: each coordinate strictly greater than i is decreased by one. Distinct such units therefore have distinct images. It follows that $\ker d_i$ is spanned by the units whose index tuples contain i. Intersecting these coordinate subspaces proves both basis statements.

For $p \geq 1$, let W_p be the span of the normalized tensor units whose indices avoid zero. The basis descriptions give

$$N_pX = Z_p(NX) \oplus W_p.$$

On W_p, the differential subtracts one from every coordinate. For $p \geq 2$, this is a bijection from its basis onto the basis of $Z_{p-1}(NX)$. Consequently every positive-degree cycle is a boundary. For $p = 1$, the image d_0W_1 is exactly the span of the degree-zero units whose index tuples contain zero. The complementary degree-zero units have $1 \leq m_a \leq c_a$ for every a. Their residue classes give the asserted basis of H_0. □

The vanishing of positive ambient homology does not force the same vanishing in a simplicial submodule. A cycle in a submodule may have all of its ambient fillings outside that submodule. The support description nevertheless imposes a bound depending only on the number of axes.

Corollary 9.2 (Bounds for submodule homology)**.** *If $Y \subseteq X(\mathbf{c})$ is a simplicial submodule with k tensor axes, then*

$$H_p(NY) = 0 \qquad (p \geq k).$$

For two axes, $\dim_K H_1(NY) \leq 2$*, independently of the tail lengths.*

Proof. For $p \geq 1$,

$$Z_p(NY) = Y_p \cap \bigcap_{i=0}^{p} \ker d_i \subseteq Z_p(NX).$$

A tuple of k coordinates cannot contain the $p+1$ distinct values $0, \dots, p$ when $p \geq k$. In degree one with two axes, the only tuples containing both zero and one are $(0,1)$ and $(1,0)$. Thus the ambient cycle space has basis E_{01}, E_{10}, and taking a subspace and then a quotient cannot increase its dimension. □

9.2. An embedding of graph chains

Let G be a finite graph with vertices $1, \dots, v$, where $v \geq 1$, and edges $1, \dots, e$. Choose an orientation of each edge, and write $u_j \to w_j$ for edge j. Loops and parallel edges are allowed. Regard G as the simplicial set with these vertices and nondegenerate edges, with no nondegenerate simplices in higher degrees. Its realization is the usual one-dimensional CW complex of the graph.

For $r \geq \max\{v, e\}$, define tensors in $X^{(3,r)}$ by

$$a_t = E_{0,t,t} \in X_0^{(3,r)} \qquad (1 \leq t \leq v), \tag{4}$$

$$T_j = E_{0,u_j+1,u_j+1} + E_{1,w_j+1,w_j+1} + E_{0,1,j+1} \in X_1^{(3,r)} \qquad (1 \leq j \leq e). \tag{5}$$

Deleting index one or zero, respectively, gives

$$d_1 T_j = a_{u_j}, \qquad d_0 T_j = a_{w_j}. \tag{6}$$

Theorem 9.3 (Graph representation)**.** *The assignments in (4)–(5) extend to an injective simplicial linear map*

$$F : K[G] \longrightarrow X^{(3,r)}.$$

Its image Y_G is the simplicial submodule generated by all a_t and T_j. Its normalized chain complex is isomorphic to

$$0 \longrightarrow K^e \xrightarrow{\partial} K^v \longrightarrow 0, \qquad \partial(\mathbf{e}_j) = \mathbf{e}_{w_j} - \mathbf{e}_{u_j}, \tag{7}$$

where $\mathbf{e}_j$ and $\mathbf{e}_t$ denote the edge and vertex basis vectors. If G has b connected components, then

$$H_0(NY_G) \cong K^b, \qquad H_1(NY_G) \cong K^{e-v+b}, \qquad H_p(NY_G) = 0 \quad (p \geq 2).$$

Proof. The endpoint identities (6) define a simplicial map from G to the underlying simplicial set of $X^{(3,r)}$; extend it linearly to obtain F. Its image is the stated generated submodule.

The normalized degree-zero map sends the vertex basis to the distinct tensor units a_t, so it is injective. In degree one, a basis of $N_1 K[G]$ is formed by $j - s_0 u_j$. Its image is

$$b_j = T_j - s_0 a_{u_j} = E_{1,w_j+1,w_j+1} - E_{1,u_j+1,u_j+1} + E_{0,1,j+1}. \tag{8}$$

The coefficient at $(0, 1, j+1)$ is one in b_j and zero in every $b_{j'}$ with $j' \neq j$. These coordinates also have zero coefficient in the endpoint terms, since those terms have first coordinate one. Hence the b_j are linearly independent, including for loops and parallel edges. The normalized map is therefore injective in degrees zero and one. It is injective in every higher degree because $N_p K[G] = 0$ there.

The natural direct-sum decomposition into degeneracies of normalized chains, Theorem 4.2, now shows that F is injective in every simplicial degree. Finally,

$$d_0 b_j = a_{w_j} - a_{u_j},$$

which identifies the normalized differential with the incidence map in (7). On each connected component, the image of this map is the subspace of vertex coefficients having total sum zero: the edge differences span this subspace along a spanning tree. The incidence map consequently has rank $v - b$, over every field. The homology formulas follow. □

Example 9.4 (The Petersen graph). For the Petersen graph, $v = 10$, $e = 15$, and $b = 1$. The construction with $r = 15$ gives a simplicial submodule of $X^{(3,15)}$ with

$$H_0 \cong K, \qquad H_1 \cong K^6, \qquad H_p = 0 \quad (p \geq 2).$$

Corollary 9.2 shows that no simplicial submodule of a two-axis DSTM can have this first homology, regardless of its rank shift.

9.3. Tensor products and reduced smash products

For tuples $\mathbf{c} = (c_1, \ldots, c_k)$ and $\mathbf{d} = (d_1, \ldots, d_\ell)$, denote their concatenation by $(\mathbf{c}, \mathbf{d})$. Tensor products of simplicial modules in this subsection are taken degreewise.

Proposition 9.5 (Concatenation of axes)**.** *There is a natural isomorphism of simplicial K-modules*

$$X(\mathbf{c}) \otimes_K X(\mathbf{d}) \cong X(\mathbf{c}, \mathbf{d}), \qquad E_m \otimes E_n \longmapsto E_{(m,n)}. \tag{9}$$

If $A \subseteq X(\mathbf{c})$ and $B \subseteq X(\mathbf{d})$ are simplicial submodules, this isomorphism restricts to an embedding $A \otimes_K B \hookrightarrow X(\mathbf{c}, \mathbf{d})$.

Proof. In every degree, the tensor-unit bases give the displayed linear isomorphism. Equation (2) shows that it commutes with every simplicial map: the same order map acts on every axis. Tensoring inclusions of vector spaces over a field gives an inclusion, so the restriction is injective in every degree. □

Different tail lengths need not be equalized for (9). If an equal-tail ambient module is preferred, there are natural zero-padding inclusions

$$X(\mathbf{c}) \hookrightarrow X(\mathbf{c}') \qquad (c_a \leq c'_a \text{ for all } a).$$

In degree p, retain the coordinates $0, \ldots, p + c_a$ on axis a and set every entry using one of its new tail coordinates to zero. The fixed-tail maps preserve old and new tails separately, so this padding commutes with faces and degeneracies. In particular,

$$X^{(k,r)} \otimes_K X^{(\ell,r)} \cong X^{(k+\ell,r)}.$$

For a pointed simplicial set S, write $\widetilde{K}[S] = K[S]/K[*]$ for its reduced free simplicial module. The smash product is the degreewise quotient $S \wedge T = (S \times T)/(S \vee T)$. An embedding of $K[S]$ also supplies an embedding of $\widetilde{K}[S]$: the latter is naturally isomorphic to the kernel of the augmentation $\varepsilon : K[S] \to K$ by

$$[\sigma] \bmod K[*] \longmapsto [\sigma] - [*].$$

Here $*$ denotes the basepoint simplex in the appropriate degree. In particular, choosing a vertex as basepoint in Theorem 9.3 yields such a reduced embedding.

Proposition 9.6 (Reduced smash representation)**.** *There is a natural isomorphism*

$$\widetilde{K}[S \wedge T] \cong \widetilde{K}[S] \otimes_K \widetilde{K}[T]. \tag{10}$$

Consequently, embeddings

$$\widetilde{K}[S] \hookrightarrow X(\mathbf{c}), \qquad \widetilde{K}[T] \hookrightarrow X(\mathbf{d})$$

induce an embedding $\widetilde{K}[S \wedge T] \hookrightarrow X(\mathbf{c}, \mathbf{d})$. *The Eilenberg–Zilber maps give a natural chain homotopy equivalence*

$$N\widetilde{K}[S \wedge T] \simeq N\widetilde{K}[S] \otimes_K N\widetilde{K}[T], \tag{11}$$

where the right side is the tensor product of chain complexes.

Proof. In each degree, a basis of either side of (10) is indexed by pairs of nonbasepoint simplices. A pair containing a basepoint represents zero on both sides. This identification commutes with all simplicial maps, including those that send a nonbasepoint simplex to the basepoint. The embedding follows from Proposition 9.5. Applying the shuffle and Alexander–Whitney maps, transported between the normalized quotient and Moore normalization, gives (11); see [4, §8.5]. □

Example 9.7 (The smash product of two circles). In the unshifted two-axis module $X^{(2,0)}$, let $a = E_{01} \in X_1^{(2,0)}$. Both faces of a are zero. The simplicial submodule A generated by a is isomorphic to $\widetilde{K}[\Delta[1]/\partial\Delta[1]]$. Indeed, sending the nondegenerate edge to a defines a map from this reduced free module; its normalized map is an isomorphism onto Ka in degree one and has zero source in every other degree. The degeneracy decomposition proves injectivity, and the generated-submodule definition proves surjectivity. Thus

$$N_1 A = Ka, \qquad N_p A = 0 \quad (p \neq 1).$$

Proposition 9.6 embeds $A \otimes_K A$ in $X^{(4,0)}$. A normalized two-cycle representing its homology generator is the shuffle

$$z = (s_1 a) \otimes (s_0 a) - (s_0 a) \otimes (s_1 a). \tag{12}$$

The simplicial identities give $d_0 z = d_1 z = d_2 z = 0$. On this cycle, the $(1,1)$-component of the Alexander–Whitney map is computed by $d_2 \otimes d_0$ and sends z to $a \otimes a$. This is the nonzero generator of the tensor product complex $NA \otimes_K NA$, which is concentrated in degree two. The Eilenberg–Zilber equivalence therefore gives

$$H_2\big(N(A \otimes_K A)\big) \cong K, \qquad H_p\big(N(A \otimes_K A)\big) = 0 \quad (p \neq 2).$$

These are the reduced homology groups of the simplicial smash product of two circles. The computation holds in characteristic two as well, where the minus sign in (12) becomes a plus sign.

9.4. Sparse computations

The accompanying Python implementation stores rational coefficients at nonzero tensor coordinates, together with the simplicial degree and the fixed tail lengths. Faces, degeneracies, normalization, and concatenation of axes are computed directly on these coordinates. Exact rational elimination on the occurring coordinates gives the following dimensions and ranks, where $\partial_p = d_0 : N_p \to N_{p-1}$:

Representation	$\dim N_0$	$\dim N_1$	$\dim N_2$	$\operatorname{rank} \partial_1$	$\operatorname{rank} \partial_2$
Petersen graph	10	15	0	9	0
Reduced circle smash	0	1	2	0	1

Both computations therefore agree with the preceding homology calculations. At shift $r = 15$, each degree-one graph tensor belongs to an ambient space with $17^3 = 4913$ coordinates but has only three nonzero entries. This sparsity does not give a general efficiency bound: duplicating one coordinate on each of k axes can replace a single entry by 2^k entries. Source code and reproducible examples are available in https://github.com/flengyel/SimplicialTensors; the experiment is `experiments/rank_shifted_homology.py`.

Appendix A. Contractibility of the Diagonal Simplicial Module

We prove that the diagonal simplicial A-module $X_\bullet(\vec{s}; A)$ is acyclic by constructing an explicit chain contraction.

Appendix A.1. The Chain Complex and Homotopy Operator

The DSTM $X_p(\vec{s}; A)$ is the A-module of functions $T : I_p \to A$. The index bounds are $M_a(p) = n_a - 1 - n + p$, where $n = \min_a(n_a) - 1$.

Remark Appendix A.1 (Index Bounds at $p = 0$). By the definition of n, there exists at least one index a_0 such that $n_{a_0} - 1 = n$. For this index, $M_{a_0}(0) = 0$. Consequently, for any index $m \in I_0$, the coordinate m_{a_0} must be 0.

Let $(X_\bullet, \partial_\bullet)$ be the associated unaugmented chain complex. We set $X_p = 0$ for $p < 0$, and $\partial_0 = 0$.

Definition Appendix A.2 (Shift-and-Truncate Homotopy H)**.** We define $H_p : X_p \to X_{p+1}$ for $p \geq -1$. Set $H_{-1} := 0$. For $p \geq 0$, $T \in X_p$, and $m \in I_{p+1}$:

$$H_p(T)(m) := \begin{cases} 0 & \text{if } \exists a : m_a = 0, \\ T(m - \vec{1}) & \text{if } \forall a : m_a > 0. \end{cases}$$

Lemma Appendix A.3. *The operator H satisfies the following identities:*

1. *For all $p \geq 0$, $d_0^{p+1} H_p = \mathrm{id}_{X_p}$.*
2. *For all $p \geq 1$ and all $i > 0$, $d_i^{p+1} H_p = H_{p-1} d_{i-1}^p$.*

Proof. Let $T \in X_p$ and $m \in I_p$.

(1) $d_0 H(T)(m) = H(T)(\Delta_0(m)) = T(m + \vec{1} - \vec{1}) = T(m)$.

(2) Let $i > 0$.

Case $p > 0$**:** If some coordinate $m_a = 0$, then $\Delta_i(m)$ also has a zero coordinate (since $i > 0$ and $\delta_i(0) = 0$), so $H_p(T)(\Delta_i(m)) = 0$ and $H_{p-1}(d_{i-1}T)(m) = 0$ by definition of H. Thus $d_i H_p(T)(m) = H_{p-1} d_{i-1}(T)(m) = 0$.

If all coordinates $m_a > 0$, then

$$d_i^{p+1} H_p(T)(m) = H_p(T)(\Delta_i^p(m)) = T(\Delta_i^p(m) - \vec{1}),$$

while

$$H_{p-1}d^p_{i-1}(T)(m) = d^p_{i-1}(T)(m-\vec{1}) = T(\Delta^p_{i-1}(m-\vec{1})).$$

The coface maps satisfy the coordinate identity

$$\delta_i(x) - 1 = \delta_{i-1}(x-1) \qquad (x > 0),$$

hence $\Delta^p_i(m) - \vec{1} = \Delta^p_{i-1}(m-\vec{1})$ when all $m_a > 0$, and the two expressions agree.

Case $p = 0$: We verify $d^1_1 H_0 = 0$.

Let $m \in I_0$. By Remark Appendix A.1, there exists a_0 such that $m_{a_0} = 0$. $d_1 H_0(T)(m) = H_0(T)(\Delta_1(m))$. The a_0-th coordinate of $\Delta_1(m)$ is $\delta_1(m_{a_0}) = \delta_1(0)$. Since $i = 1 > 0$, $\delta_1(0) = 0$. By the definition of H, since $\Delta_1(m)$ has a zero coordinate, $H_0(T)(\Delta_1(m)) = 0$. Thus $d^1_1 H_0 = 0$. □

Theorem Appendix A.4. *The diagonal simplicial module $X_\bullet(\vec{s}; A)$ is contractible.*

Proof. We verify the chain contraction identity

$$\partial_{p+1}H_p + H_{p-1}\partial_p = \mathrm{id}_{X_p} \qquad (p \geq 0).$$

Base case $p = 0$. Here $\partial_0 = 0$ and $H_{-1} = 0$. Also $\partial_1 = d^1_0 - d^1_1$. Thus

$$\partial_1 H_0 = (d^1_0 - d^1_1)H_0 = d^1_0 H_0 - d^1_1 H_0 = \mathrm{id}_{X_0} - 0 = \mathrm{id}_{X_0},$$

using Lemma Appendix A.3(1) for $p = 0$ and Lemma Appendix A.3(2) in the special $p = 0$ verification (i.e. $d^1_1 H_0 = 0$). Therefore

$$\partial_1 H_0 + H_{-1}\partial_0 = \mathrm{id}_{X_0}.$$

Inductive case $p \geq 1$. Using Lemma Appendix A.3:

$$\begin{aligned}\partial_{p+1}H_p &= d^{p+1}_0 H_p + \sum_{i=1}^{p+1}(-1)^i d^{p+1}_i H_p \\ &= \mathrm{id}_{X_p} + \sum_{i=1}^{p+1}(-1)^i H_{p-1}d^p_{i-1}.\end{aligned}$$

Re-indexing with $j = i - 1$ gives

$$\begin{aligned}\sum_{i=1}^{p+1}(-1)^i H_{p-1}d^p_{i-1} &= \sum_{j=0}^{p}(-1)^{j+1} H_{p-1}d^p_j \\ &= -H_{p-1}\partial_p.\end{aligned}$$

Substituting back yields

$$\partial_{p+1}H_p + H_{p-1}\partial_p = \mathrm{id}_{X_p}.$$

Thus H is a chain contraction of $X_\bullet(\vec{s}; A)$. □

Appendix A.2. Equivariance Properties of the Homotopy

Recall the action of the stabilizer group $\mathrm{Stab}(\vec{s})$ on the diagonal simplicial module $X_\bullet(\vec{s}; A)$ defined in Section 2. The face and degeneracy maps commute with this action (Lemma 2.5).

Lemma Appendix A.5 (Equivariance under $\mathrm{Stab}(\vec{s})$)**.** *For every $p \geq -1$ and $\sigma \in \mathrm{Stab}(\vec{s})$, the shift-and-truncate homotopy H_p is equivariant with respect to the action of $\mathrm{Stab}(\vec{s})$:*

$$H_p(\sigma \cdot T) = \sigma \cdot H_p(T).$$

Proof. Let $T \in X_p$, $\sigma \in \mathrm{Stab}(\vec{s})$, and $m \in I_{p+1}$. We compare $H_p(\sigma \cdot T)(m)$ and $(\sigma \cdot H_p(T))(m)$. By definition of the action, $(\sigma \cdot H_p(T))(m) = H_p(T)(\sigma^{-1}m)$.

The definition of H_p depends on the presence of zero coordinates.

Case 1: $\exists a : m_a = 0$. In this case, $H_p(\sigma \cdot T)(m) = 0$. Since $\sigma^{-1}m$ also contains a zero coordinate, $H_p(T)(\sigma^{-1}m) = 0$.

Case 2: $\forall a : m_a > 0$. In this case, $\sigma^{-1}m$ is also strictly positive.

$$H_p(\sigma \cdot T)(m) = (\sigma \cdot T)(m - \vec{1}) = T(\sigma^{-1}(m - \vec{1})).$$

On the other hand,

$$(\sigma \cdot H_p(T))(m) = H_p(T)(\sigma^{-1}m) = T((\sigma^{-1}m) - \vec{1}).$$

The equality holds because the shift operation commutes with permutation: $\sigma^{-1}(m - \vec{1}) = (\sigma^{-1}m) - \vec{1}$. Therefore, H_p is equivariant under the action of $\mathrm{Stab}(\vec{s})$. □

Appendix A.3. Shifted Depth Filtration and Spectral Sequence

We introduce a filtration on the complex $(X_\bullet, \partial)$ that is compatible with the differential and the homotopy operator H, leading to the collapse of the associated spectral sequence.

Definition Appendix A.6 (Shifted Depth Filtration)**.** For $p \geq 0$ and $t \in \mathbb{Z}$, define a decreasing filtration $\mathcal{F}^\bullet X_p$. Write $\min(m) := \min_a m_a$ for $m \in I_p$. Set

$$\mathcal{F}^t X_p \;:=\; \{\, T \in X_p \mid T(m) = 0 \text{ unless } \min(m) \geq t + p \,\}.$$

This filtration is bounded: $\mathcal{F}^{-p} X_p = X_p$ (since $\min(m) \geq 0$). Furthermore, since $\min(m) \leq \min_a M_a(p)$ for any $m \in I_p$, we have $\mathcal{F}^t X_p = 0$ if $t + p > \min_a M_a(p)$.

Lemma Appendix A.7 (Behavior of ∂ and H w.r.t. $\mathcal{F}^\bullet$)**.** *The differential ∂ and the homotopy H preserve the shifted depth filtration (filtration degree 0). For all $t \in \mathbb{Z}$:*

1. *For all $p \geq 1$ and all $i \in \{0, \ldots, p\}$,*
$$d_i^p(\mathcal{F}^t X_p) \subseteq \mathcal{F}^t X_{p-1}.$$
2. *For all $p \geq 0$,*
$$H_p(\mathcal{F}^t X_p) \subseteq \mathcal{F}^t X_{p+1}.$$

Proof. (1) Let $p \geq 1$ and let $T \in \mathcal{F}^t X_p$. We show $d_i^p(T) \in \mathcal{F}^t X_{p-1}$. Let $m \in I_{p-1}$. If $\min(m) < t+p-1$, then

$$d_i^p(T)(m) = T(\Delta_i^p(m)).$$

For each coordinate, the coface map satisfies $\delta_i(x) \leq x+1$, hence

$$\min(\Delta_i^p(m)) \leq \min(m) + 1.$$

If $\min(m) < t+p-1$, then $\min(\Delta_i^p(m)) < t+p$, and since $T \in \mathcal{F}^t X_p$ we have $T(\Delta_i^p(m)) = 0$. Therefore $d_i^p(T) \in \mathcal{F}^t X_{p-1}$.

(2) Let $p \geq 0$ and let $T \in \mathcal{F}^t X_p$. We show $H_p(T) \in \mathcal{F}^t X_{p+1}$. Let $m \in I_{p+1}$. If $\min(m) = 0$, then $H_p(T)(m) = 0$ by definition of H. If $\min(m) > 0$, then $H_p(T)(m) = T(m - \vec{1})$. If $\min(m) < t+p+1$, then

$$\min(m - \vec{1}) = \min(m) - 1 < t + p,$$

so $T(m - \vec{1}) = 0$ because $T \in \mathcal{F}^t X_p$. Hence $H_p(T) \in \mathcal{F}^t X_{p+1}$. □

Proposition Appendix A.8 (Spectral sequence collapses at E_1)**.** *Let $\mathcal{F}^\bullet$ be the shifted depth filtration. The spectral sequence of the filtered complex $(X_\bullet, \partial, \mathcal{F}^\bullet)$ satisfies*

$$E_1^{t,q} = 0 \qquad \textit{for all } t, q.$$

Hence it collapses at the E_1-page.

Proof. By Lemma Appendix A.7, for $p \geq 1$ the differential ∂_p has filtration degree 0, and since $\partial_0 = 0$ the filtered complex $(X_\bullet, \partial, \mathcal{F}^\bullet)$ is well-defined.

Let $E_0 = \mathrm{gr}_{\mathcal{F}}(X_\bullet)$ be the associated graded complex, with differential d^0 induced by ∂.

By Lemma Appendix A.7, the homotopy operator H also has filtration degree 0. It induces a map $[H]$ on E_0. The chain contraction identity $\partial H + H\partial = \mathrm{id}$ holds in the filtered complex. Since all operators involved preserve the filtration, the identity descends to the associated graded complex:

$$d^0[H] + [H]d^0 = [\mathrm{id}] = \mathrm{id}_{E_0}.$$

This shows that the complex (E_0, d^0) is contractible via the homotopy $[H]$. Therefore, its homology is trivial. The E_1 term is defined as the homology of (E_0, d^0). We conclude $E_1 = H(E_0) = 0$. □

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